PROTECTED PLANAR CARRIERS: SEPARATION LEDGERS AND ACCESS-PRESERVING ROUTING

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Abstract. We give a finite planar theory for joining pairwise disjoint full continua while preserving specified prime-end accesses and avoiding a frozen compact inventory. A protected ribbon-routing certificate is realized by a full carrier whose null crosscut chains survive, remain distinct when their relative collar tails are disjoint, and retain cyclic order, without any local-connectivity hypothesis. Certificate existence is equivalent to one component of the frozen free domain admitting simultaneous singleton ports to all atoms. Packet and cellwise criteria give checkable sufficient conditions, while fixed-assignment audits give persistent witnesses for merger, escape, and port loss.

We also prove an attachment exact sequence. If a connected connector C meets a full atom A in T, then the new first Čech cohomology fits into the short exact sequence $0 \to \widetilde{\check{H}}^{\,0}(T) \to \check{H}^1(C \cup A) \to \ker[\check{H}^1(C) \to \check{H}^1(T)] \to 0$. Iteration gives the separation ledger

$$\sigma\left(C \cup \bigcup_i A_i\right) = \sigma(C) + \sum_i \big(\#\pi_0(C \cap A_i) - 1 - \sigma(C \cap A_i)\big)$$

whenever the ranks are finite. Thus disconnected traces create holes and separating traces fill pre-existing holes. For a full connector the union is full exactly when every attachment trace is connected. Arbitrarily many holes can nevertheless be born in a Hausdorff limit of full tree carriers with singleton traces. By contrast, decreasing full connector envelopes with nonempty connected traces are stable and preserve common protected data. These results isolate the precise finite/infinite compactness gate; they make no claim of constructing a virtual dynamical carrier or proving Mandelbrot local connectivity.

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1. Introduction

1.1. **The routing problem.** Let $s \geq 1$, and let $A_1, \dots, A_s$ be pairwise disjoint full continua in a Jordan disk $U \subset \mathbb{C}$. The elementary task of joining the A_i becomes substantially less elementary when the connector must also avoid a frozen compact obstacle set and preserve designated prime ends of the complements $\widehat{\mathbb{C}} \setminus A_i$. A connector can erase such an access by entering its selected side, identify two accesses that were distinct before enlargement, or disconnect the region through which later ports were meant to run. None of these failures is detected by connectedness of the final union.

This problem appears naturally when small filled Julia sets, puzzle pieces, or combinatorial fibers must be organized without assuming local connectivity. Polynomial-like maps provide the basic continua [2]; Kahn–Lyubich impose finite-decoration combinatorics and prove a priori bounds for that class [8]; recent work on primitive renormalization constructs native or model-lifted Hubbard trees when the required dynamical structure is available [7]; and Dudko's roadmap emphasizes connected hulls of families of small Julia sets as one ingredient in the broader local-connectivity program [4]. The present paper isolates the planar problem that remains after the dynamical data have selected finitely many continua and accesses but before any canonical connector has been justified.

The hypotheses deliberately allow non-locally-connected boundaries. A protected access is therefore not represented by a presumed landing ray or boundary parametrization. It is represented by a null prime-end crosscut chain together with its selected sides. The part forbidden to a future connector is the *relative collar tail* $\overline{D} \setminus A_i$, not the whole closed side $\overline{D}$. This distinction is essential: different prime ends may have overlapping impressions inside the same atom even though their relative collar tails in the complement are disjoint.

1.2. **Main results.** The first conclusion is negative but foundational. There is no least compact connected full hull even for two points (Theorem 3.1), so "take the smallest carrier" is not a definition. Nor can one safely choose a carrier and absorb pulled-back excess monotonically: Proposition 3.2 gives a degree-one conformal branch, with no critical-value obstruction, in which the first absorption erases a protected access permanently. These examples show that the isotopy/access class of the routing is additional data.

The positive replacement is a *protected ribbon-routing certificate*. It records free singleton ports from the atoms, a finite embedded tree joining their outer endpoints, the induced ribbon order, every protected crosscut chain, and the complete frozen inventory to be avoided. Theorem 5.4 realizes any such certificate by a compact connected full carrier. Every protected chain remains a prime-end chain of the enlarged complement, distinct protected chains remain distinct whenever their relative collar tails are disjoint, and cyclic order is preserved within each common source complement. No local connectivity of an atom is used.

The first structural result developed here is an exact attachment ledger. For a compact connected connector C, full atoms A_i, and traces $T_i = C \cap A_i$, Theorem 6.1 extracts from the compact-pair Mayer–Vietoris sequence a natural short exact sequence at every attachment. When the relevant component counts are finite, it gives

$$\sigma\left(C \cup \bigcup_i A_i\right) = \sigma(C) + \sum_i \big(\#\pi_0(T_i) - 1 - \sigma(T_i)\big),$$

where $\sigma(K) = \#\pi_0(\widehat{\mathbb{C}} \setminus K) - 1$. The formula is a credit/debit law. Each extra component of an attachment trace creates a complementary component; each complementary component of the trace fills one separation already present in the connector. In particular, if C and the atoms are full, every T_i is nonseparating and the union is full if and only if all traces are connected (Corollary 6.3). Thus the familiar tree with singleton attachments is only one sufficient model: an arbitrary full continuum connector with arbitrary connected traces works just as well.

The same ledger identifies a compactness gate for this framework. Proposition 6.5 constructs, for every m, full arc connectors C_n meeting an atom once and converging in the Hausdorff metric to a full arc C whose trace has $m + 1$ components. Each finite-stage union $A \cup C_n$ is full, while the limit union has exactly m bounded complementary components. Connector cycles can likewise be born in a limit. Consequently neither fullness nor the singleton-trace condition passes to a virtual or infinite carrier by Hausdorff compactness alone: a limiting argument must separately control connector cohomology and trace multiplication (Warning 6.7). Theorem 6.6 gives a positive counterpart: a decreasing sequence of full connector envelopes with connected nonempty traces has a full limiting connector with connected traces, and common protected data survive. This identifies a concrete inverse-limit architecture that avoids both defect-birth mechanisms.

The second structural result is an exact routing criterion. Freeze the atoms, relative protected collar tails, obstacles, critical exclusions, and localization-forbidden data into a compact set Q, and put $\Omega_{\mathrm{free}} = U \setminus Q$. Theorem 7.3 says that a protected certificate exists if and only if one component of Ω_{free} admits simultaneous singleton ports to every required atom. If no component of this fixed free domain has simultaneous ports,

the failure persists under every connector enlargement respecting the same Q. The proof uses a one-ended slit lemma whose statement keeps the free tips inside the residual domain. The source version deleted those tips and mapped a boundary-anchored slit into an open domain; the corrected Lemma 7.1 repairs both defects, and the proof of the criterion is rebuilt around the repaired statement.

Finally, the exact criterion is converted into finite audits. Theorem 8.4 packages the frozen inventory into pairwise disjoint full continua and uses compacta Alexander duality to force a common free domain. Theorem 9.3 proves that fullification is independent of processing order for a fixed typed assignment and reduces success to coverage, packet separation/localization, and a nonempty unprotected exposure locus at each atom. Packet merger, localization escape, and port loss come with witnesses persistent under every same-label monotone enlargement. Proposition 9.5 supplies a directly checkable Jordan-cell criterion.

1.3. **Position.** The underlying tools are classical: Janiszewski–Mullikin separation [9, 11, 13, 17], the classical theories of unicoherence and multicoherence [1, 5], Carathéodory prime ends [14], the Jordan–Schoenflies theorem, filled hulls, and compacta Alexander duality and Mayer–Vietoris [15, Chapter 6]. We do not claim a new foundation for planar topology. The contribution is a theorem-level synthesis aimed at a specific proof obligation that the standard constructions do not record: finite connection with named access survival, exact existence and failure tests, and a quantitative accounting of the separations created or destroyed by attachment.

In particular, the separation-ledger theorem is formally a consequence of classical duality and Mayer–Vietoris; its full/full zero-defect criterion is part of the Janiszewski–Mullikin tradition. Its value here is the homomorphism-level diagnostic, the creation-versus-cancellation formula, and the limit obstruction it exposes. The other central statements turn prime-end intentions into finite data: null chains and relative collar tails, singleton ports, an embedded ribbon tree, a frozen free domain, typed packet assignments, and persistent witnesses. Section 4 compares this package with least hulls, monotone absorption, regulated and Hubbard trees, Moore quotients, and a lamination-lift candidate.

This architecture is intentionally distinct from the companion direct-decoration manuscript [16]. That manuscript contains the rooted-end compactification, partial-shift dynamics, gap cells, arbitrary packet replacement, and the finite decoration-carrier application. The present paper contains the general planar certification theory, the exact free-component and packet criteria, the attachment ledger, and the Hausdorff-limit no-go theorem. Separating the two removes a long duplicated application and makes the logical dependency transparent.

1.4. **Scope and organization.** Everything here is finite and planar. The paper does not construct a virtual carrier or “Wave horizon” (the source

manuscript's proposed infinite limiting carrier), an analytic correspondence, a uniform incidence bound, or a compression mechanism, and it proves no new case of local connectivity of the Mandelbrot set. The recent theorem of Kahn–Kapiamba–Lyubich applies in its stated parabolically bounded primitive category [7]; it is a positive benchmark, not a consequence of the present framework. Our conclusion is conditional in exactly one transparent sense: once finite protected data or a valid packet assignment are supplied, the planar realization problem is solved, while the production of coherent data at an infinite dynamical horizon remains open.

Section 2 fixes the compacta and prime-end conventions. Section 3 proves the two motivating obstructions, and Section 4 places the framework among standard constructions. Section 5 gives certificate realization and prime-end survival. Section 6 proves the attachment ledger and its sharp Hausdorff-limit obstruction. Section 7 proves the corrected slit lemma and exact free-component criterion. Sections 8 and 9 develop the packet criterion, order-independent fullification, persistent audits, and the cellwise test. Section 10 closes with the exact interface to future finite and limiting applications.

2. Notation and classical background

Everything in this section is classical. It is collected once, in the normalized forms used by the rest of the paper, for three reasons: the statements are invoked repeatedly and in exact form; two of them — the compacta form of Alexander duality and the tameness of plane arcs — are easy to quote in weaker versions that would not support the proofs; and the source manuscript's terminology — *atom*, *package*, *carrier*, *fullification*, *protected access*, *collar* — is fixed here at first use.

2.1. **Plane, sphere, and general notation.** Throughout, $\mathbb{C}$ is the complex plane, $\mathbb{Z}$ the integers, and $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ the Riemann sphere, identified with the round sphere S^2 by stereographic projection and carrying its standard orientation. Compact subsets of $\mathbb{C}$ are exactly the compact subsets of $\widehat{\mathbb{C}}$ missing ∞, and we pass between the two ambient spaces without comment. The symbol dist denotes Euclidean distance in $\mathbb{C}$; diameters of crosscuts (Section 2.4) are measured in the spherical metric, and all other metric quantities are Euclidean. For a set X, $\operatorname{int} X$ is its interior, ∂X its boundary, and $\overline{X}$ its closure, taken in the ambient space clear from context; the relative closure of X in a subspace Y is written $\overline{X}^{Y}$, as in the attachment trace (8).

For sets A, B we write $A \Subset B$, and say A is *compactly contained* in B, if $\overline{A}$ is compact and $\overline{A} \subset \operatorname{int} B$; this is the manuscript's relation $\Subset$. A *domain* is a nonempty connected open subset of $\mathbb{C}$ or $\widehat{\mathbb{C}}$; a *plane domain* is a domain in $\mathbb{C}$. Every plane domain is polygonally connected: any two of its points are joined by a polygonal path inside it [13]. A *Jordan curve* is a homeomorphic image of the circle, a *Jordan arc* a homeomorphic image of $[0, 1]$, and an *open Jordan arc* a Jordan arc with its two endpoints removed.

A *Jordan disk* (synonymously, a *Jordan domain*) is a bounded plane domain whose boundary is a Jordan curve; by the Jordan and Schoenflies theorems recalled in Proposition 2.8 below, a Jordan disk is simply connected, and its closure — a *closed Jordan disk* — is homeomorphic to the closed unit disk. A *continuum* is a nonempty compact connected subset of $\mathbb{C}$ or $\widehat{\mathbb{C}}$; it is *nondegenerate* if it contains more than one point.

2.2. **Full continua, filled hulls, and fullification.** A compact set $K \subset \mathbb{C}$ is *full* if $\mathbb{C} \setminus K$ is connected. Following the source manuscript (equation (9.5) of [12]), the *filled hull* of a compact set $K \subset \mathbb{C}$ is

$$\mathrm{Fill}(K) \;=\; K \cup \bigcup \big\{\, B : B \text{ a bounded component of } \mathbb{C} \setminus K \,\big\}, \qquad (1)$$

that is, K together with the bounded components of its complement. The passage from K to $\mathrm{Fill}(K)$ is called *fullification*; the term is the manuscript's, and it is used as a noun for the operation and, occasionally, for its result. The following properties are used constantly and are recorded with proof.

Lemma 2.1 (Basic properties of filled hulls)**.** *Let $K \subset \mathbb{C}$ be compact, and let $\Omega_\infty(K)$ denote the unbounded component of $\mathbb{C} \setminus K$.*

1. $\mathrm{Fill}(K) = \mathbb{C} \setminus \Omega_\infty(K)$*; in particular* $\mathrm{Fill}(K)$ *is compact, and it is contained in every closed disk containing K.*
2. *The following are equivalent: K is full;* $\widehat{\mathbb{C}} \setminus K$ *is connected;* $\mathrm{Fill}(K) = K$*.*
3. *If K is a continuum, then* $\mathrm{Fill}(K)$ *is a full continuum, and* $\widehat{\mathbb{C}} \setminus \mathrm{Fill}(K) = \Omega_\infty(K) \cup \{\infty\}$ *is a simply connected domain containing ∞.*
4. *The operator* Fill *is extensive ($K \subset \mathrm{Fill}(K)$), monotone ($K \subset L$ implies $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$), and idempotent ($\mathrm{Fill}(\mathrm{Fill}(K)) = \mathrm{Fill}(K)$).*

Proof. (1) The set $\mathbb{C} \setminus \Omega_\infty(K)$ consists of K together with all bounded components of $\mathbb{C} \setminus K$, which is (1); it is closed because $\Omega_\infty(K)$ is open. If E is a closed disk with $K \subset E$, then $\mathbb{C} \setminus E$ is a connected unbounded subset of $\mathbb{C} \setminus K$, hence contained in $\Omega_\infty(K)$; taking complements gives $\mathrm{Fill}(K) \subset E$. Thus $\mathrm{Fill}(K)$ is closed and bounded.

(2) The components of $\widehat{\mathbb{C}} \setminus K$ are the bounded components of $\mathbb{C} \setminus K$ together with $\Omega_\infty(K) \cup \{\infty\}$: the latter set is open in $\widehat{\mathbb{C}}$ (a spherical neighborhood of ∞ misses the compact set K, and its intersection with $\mathbb{C}$, being connected and unbounded, lies in $\Omega_\infty(K)$), it is connected, and its complement within $\widehat{\mathbb{C}} \setminus K$ is the union of the bounded components, which is open. Hence $\widehat{\mathbb{C}} \setminus K$ is connected exactly when there are no bounded components, which is fullness of K, which by (1) is the identity $\mathrm{Fill}(K) = K$.

(3) Let K be a continuum and let B be a bounded component of $\mathbb{C} \setminus K$. The boundary of B is contained in K: if a boundary point x of B lay in $\mathbb{C} \setminus K$, its component B' in $\mathbb{C} \setminus K$ would be an open set containing x, hence meeting B, hence equal to B; but then $x \in B$, contradicting openness of B at its own boundary point. So $\partial B \subset K$. Thus $\overline{B} = B \cup \partial B$ is a connected

set meeting K, and $\mathrm{Fill}(K) = K \cup \bigcup_B \overline{B}$ is a union of connected sets each meeting the connected set K, hence connected. With (1) this makes $\mathrm{Fill}(K)$ a continuum, and its complement in $\widehat{\mathbb{C}}$ is the domain $\Omega_\infty(K) \cup \{\infty\}$, so $\mathrm{Fill}(K)$ is full. Finally, a domain in $\widehat{\mathbb{C}}$ whose complement is connected is simply connected [13, 17], and the complement of the domain $\Omega_\infty(K) \cup \{\infty\}$ is the continuum $\mathrm{Fill}(K)$.

(4) Extensivity is (1). For monotonicity, let $K \subset L$; then $\Omega_\infty(L)$ is a connected unbounded subset of $\mathbb{C} \setminus K$, hence $\Omega_\infty(L) \subset \Omega_\infty(K)$, and taking complements in (1) gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$. For idempotence, the complement of $\mathrm{Fill}(K)$ is the single unbounded component $\Omega_\infty(K)$, so $\mathrm{Fill}(\mathrm{Fill}(K)) = \mathbb{C} \setminus \Omega_\infty(K) = \mathrm{Fill}(K)$. □

The deeper absorption law $\mathrm{Fill}(\mathrm{Fill}(K) \cup L) = \mathrm{Fill}(K \cup L)$, which makes fullification of a typed packet independent of processing order, is Lemma 9.1 below (manuscript Lemma 10.12); it is a formal consequence of the three properties in clause (4).

Definition 2.2 (Package, atoms, carrier, fullification)**.** A *package* is a nonempty finite union $S = A_1 \cup \cdots \cup A_s$, $s \geq 1$, of pairwise disjoint full continua, called its *atoms*. Atoms may be degenerate: in Theorem 3.1 the package consists of the two points ± 1. A *carrier* of the package S is a compact connected full continuum $H \subset \mathbb{C}$ with $S \subset H$. Both *atom* — a required full continuum of the package — and *fullification* — the passage (1) — are the source manuscript's terms [12] and are used in its sense throughout.

2.3. **Janiszewski's separation theorem.** A closed set $F \subset \widehat{\mathbb{C}}$ *separates* points $p, q \in \widehat{\mathbb{C}} \setminus F$ if p and q lie in different components of $\widehat{\mathbb{C}} \setminus F$; the set F is *nonseparating* if $\widehat{\mathbb{C}} \setminus F$ is connected. By Lemma 2.1(2), a compact subset of $\mathbb{C}$ is nonseparating in $\widehat{\mathbb{C}}$ exactly when it is full; the two words are used interchangeably for such sets.

Theorem 2.3 (Janiszewski's separation theorem; [9]; see also [13, 17])**.** *Let $F_1, F_2 \subset \widehat{\mathbb{C}}$ be closed sets whose intersection $F_1 \cap F_2$ is connected and nonempty, and let $p, q \in \widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$. If neither F_1 nor F_2 separates p from q, then $F_1 \cup F_2$ does not separate p from q.*

Corollary 2.4. *If $F_1, F_2 \subset \widehat{\mathbb{C}}$ are closed and nonseparating and $F_1 \cap F_2$ is connected and nonempty, then $F_1 \cup F_2$ is nonseparating. In particular, the union of two full compact subsets of $\mathbb{C}$ whose intersection is connected and nonempty is full.*

Proof. Let $p, q \in \widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$. Since $\widehat{\mathbb{C}} \setminus F_1$ is connected, F_1 does not separate p from q, and likewise for F_2. By Theorem 2.3, $F_1 \cup F_2$ does not separate p from q. As the pair was arbitrary, $\widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$ is connected. The final sentence follows from Lemma 2.1(2). □

This is the form used in the Janiszewski induction of Lemma 5.2, where the intersections are singletons. The complementary situation of finitely many pairwise *disjoint* full compacta, where the stated form of the theorem does not apply, is handled cohomologically, by the duality of Section 2.5 and the nerve argument of Lemma 8.3 (manuscript Lemma 10.10).

2.4. **Prime ends, crosscut chains, and protected accesses.** Informally, an *access* to a compact set K is a way of reaching ∂K from the complement. It can be presented by a landing arc, as the access $p_0^+(m)$ of Proposition 3.2 is presented by the vertical segment $m+it$; but landing presentations presuppose that something lands, and for the arbitrary continua handled in this paper a landing presentation need not be available. The presentation that requires nothing is Carathéodory's theory of prime ends [14], which we now recall in the normalization used throughout.

Let $\Omega \subset \widehat{\mathbb{C}}$ be a simply connected domain whose complement contains at least two points. The standing example is $\Omega = \widehat{\mathbb{C}} \setminus K$ for a nondegenerate full continuum K, in which case Ω is a simply connected domain containing ∞ by Lemma 2.1(3); Jordan disks are a second example. A *crosscut* of Ω is an open Jordan arc $\gamma \subset \Omega$ whose closure $\overline{\gamma}$ is a Jordan arc with two distinct endpoints on $\partial\Omega$. A crosscut divides Ω into exactly two components, each a simply connected domain having γ on its boundary [14].

A *crosscut chain* (or *prime-end chain*; Pommerenke's null chains [14], in the following normalization) is a sequence

$$\Gamma = (\gamma_n, D_n)_{n\geq 1}$$

in which the γ_n are pairwise disjoint crosscuts of Ω, $\overline{\gamma_n}$ are pairwise disjoint, D_n is one of the two components of $\Omega \setminus \gamma_n$, and

$$\overline{D_{n+1}} \cap \Omega \subset D_n, \qquad \operatorname{diam}_{\mathrm{sph}}(\gamma_n) \longrightarrow 0. \tag{2}$$

The inclusion is the strict nesting condition, while diameter convergence is the null condition. The strict nesting implies both $D_{n+1} \subset D_n$ and $\gamma_{n+1} \subset \overline{D_{n+1}} \cap \Omega \subset D_n$; thus a later crosscut lies strictly on the declared side of every earlier one. This stronger formulation rules out the large-side pathologies that mere nestedness of the open domains would permit. The record of a chain thus contains strictly more than the crosscuts: it contains, for every n, the declaration of the selected side, and it is exactly this declaration that the certificates of Section 5 protect. Two chains are *equivalent* if each eventually divides the other, that is, if every selected domain of either chain contains all sufficiently deep selected domains of the other. An equivalence class is a *prime end* of Ω.

Two classical facts about prime ends are used. First, the Carathéodory correspondence: an orientation-preserving Riemann map $\psi\colon \{|z|<1\} \to \Omega$ induces a bijection between the points of the unit circle and the prime ends of Ω; radial approach converges in the prime-end compactification, although a Euclidean radial limit need not exist without additional boundary regularity;

see [14, Chapter 2]. Second, the *impression*

$$I(\Gamma) \;=\; \bigcap_{n\geq 1} \overline{D_n}$$

(closures in $\widehat{\mathbb{C}}$) is a nonempty compact connected subset of $\partial\Omega$ — the intersection of a decreasing sequence of nonempty compact connected sets — and depends only on the prime end of Γ, not on the representing chain [14]. The nested closures $\overline{D_n}$ decrease *to* the impression; this is the form in which impressions enter later collar and exposure arguments.

The prime ends of Ω carry a natural *cyclic order*: the one transported from the unit circle by the Carathéodory correspondence. It is independent of the choice of Riemann map, since two orientation-preserving Riemann maps differ by a disk automorphism, which preserves the cyclic orientation of the circle. For a finite family of prime ends represented by chains with pairwise disjoint selected sides, the cyclic order can be computed inside the domain, with no reference to a uniformization: choose one sufficiently deep oriented crosscut from each chain; the cyclic order of the prime ends is the separation order of these pairwise disjoint crosscuts in Ω. The computation is carried out carefully, by means of embedded triods, in the proof of Lemma 5.3, and that formulation is the one shown to be invariant under carrier enlargement.

Definition 2.5 (Protected access; collar)**.** Let A be a nondegenerate atom. A *protected access* to A is a distinguished null prime-end crosscut chain $\Gamma = (\gamma_n, D_n)_{n\geq 1}$ of $\widehat{\mathbb{C}} \setminus A$, recorded together with its selected sides. The terminal selected domain D_1 is the *collar* of the access, and

$$\overline{D_1} \setminus A$$

is its *relative collar tail*. When finitely many protected accesses appear together inside a Jordan localization, their labels are required to specify pairwise distinct prime ends at each fixed atom, and the chains are *truncated*—an initial segment of each chain is discarded and the indices shifted—so that the relative terminal collar tails are pairwise disjoint, lie in the localization, and miss the forbidden set and the other atoms. For chains based at one atom, disjoint sufficiently deep tails come from Hausdorff separation in that atom's prime-end closed disk. For chains based at different atoms, and for avoidance of the forbidden compactum, the same conclusion follows separately from compact separation together with the fact that the nested compact sets $\overline{D_{i,a,n}}$ decrease to the impression in A_i. Their closures may meet inside a common atom: distinct prime ends can have overlapping impressions, and no local connectivity assumption is being smuggled into the normalization. Section 5 records this convention as part of the certificate data.

The adjective *protected* names a discipline, not a topological property: a protected access is one that every later construction is forbidden to disturb, in the precise sense that no connector may enter its relative collar tail (Definition 5.1). The reward for the discipline is the survival theorem,

Lemma 5.3: the chain then persists verbatim, with the same crosscuts and the same selected sides, as a prime-end chain of the enlarged complement.

2.5. **Čech cohomology and Alexander duality.** For a compact set $K \subset S^2$, $\check{H}^*(K;\mathbb{Z})$ denotes Čech cohomology with integer coefficients. Two of its properties are needed [15, Chapter 6]: continuity — the Čech cohomology of K is the direct limit of the singular cohomology of the open neighborhoods of K, so that every class and every relation is realized in some neighborhood — and the Mayer–Vietoris sequence for pairs of compact sets in S^2. Both are used in the nerve computation of Lemma 8.3 and in the second proof of Lemma 5.2. The duality statement is the following.

Theorem 2.6 (Alexander duality, compacta form; [15, Chapter 6])**.** *For every nonempty proper compact subset $K \subset S^2$ and every integer q, there is a natural isomorphism*

$$\widetilde{\check{H}}^{\,q}(K;\mathbb{Z}) \;\cong\; \widetilde{H}_{1-q}(S^2 \setminus K;\mathbb{Z}).$$

In particular,

$$\widetilde{H}_0\big(S^2 \setminus K;\mathbb{Z}\big) \;\cong\; \check{H}^1(K;\mathbb{Z}). \tag{3}$$

For $q = 2$, the same theorem gives $\check{H}^2(K;\mathbb{Z}) = 0$, since $S^2 \setminus K$ is nonempty and its reduced homology in degree -1 vanishes.

The general statement above is the Čech-cohomology form for compacta given by Spanier. Hatcher's Corollary 3.45 gives the familiar locally contractible special case [6, pp. 255–257].

This is equation (9.6) of the manuscript [12]. Reading (3) at the level of connectivity gives the dictionary used throughout the paper. The complement is open in a surface and hence locally path connected, so its connected and path components agree and singular H_0 detects connectedness. Thus $S^2 \setminus K$ is connected exactly when $\check{H}^1(K;\mathbb{Z}) = 0$, and hence, by Lemma 2.1(2), a compact nonempty set $K \subset \mathbb{C}$ is full exactly when $\check{H}^1(K;\mathbb{Z}) = 0$. "Nonseparating" and "vanishing first Čech cohomology" are synonyms, and the two proofs of Lemma 5.2 translate one into the other.

Warning 2.7. The Čech form of (3), valid for an arbitrary nonempty proper compact $K \subset S^2$, is required; the locally contractible special case, in which Čech and singular cohomology agree [6], is not sufficient. The atoms of this paper are arbitrary full continua—in general not locally connected or locally contractible, and not necessarily of the homotopy type of CW complexes—and for such compacta duality genuinely fails with singular cohomology in place of Čech. Hatcher's standard quasi-circle (Warsaw circle) $W \subset \mathbb{C}$, obtained by completing the closed topologist's sine curve by an arc, is path connected and simply connected but has exactly two complementary components [6, Exercise 1.3.7 and pp. 255–256]. Thus $\widetilde{H}_0(S^2 \setminus W;\mathbb{Z}) \cong \mathbb{Z}$ and compacta duality gives $\check{H}^1(W;\mathbb{Z}) \cong \mathbb{Z}$, whereas $H_1(W;\mathbb{Z}) = 0$ and the universal coefficient theorem gives $H^1_{\mathrm{sing}}(W;\mathbb{Z}) = 0$. Every duality argument

in this paper — including Lemma 5.2 (second proof), Theorem 6.1, and Lemma 8.3 — is therefore carried out in Čech cohomology, and the nerve and continuity properties invoked there are the Čech ones.

2.6. **Jordan, Schoenflies, and Carathéodory extensions.** The naturality statements of this paper (Proposition 5.6, the naturality clauses of Theorem 9.3) rest on a package of classical extension theorems, collected here with the exact attributes used later.

Proposition 2.8 (Extension package)**.**

(1) (Jordan; Schoenflies) *Every Jordan curve $J \subset \widehat{\mathbb{C}}$ divides $\widehat{\mathbb{C}}$ into exactly two components, each a simply connected domain with boundary J, and the closure of each is homeomorphic to the closed unit disk. Moreover, every homeomorphism between two Jordan curves in $\widehat{\mathbb{C}}$ extends to a homeomorphism of $\widehat{\mathbb{C}}$.*

(2) (Tameness of plane arcs) *For every Jordan arc $\alpha \subset \mathbb{C}$ there is a homeomorphism of $\mathbb{C}$ carrying α onto the straight segment $[0,1]\times\{0\}$. In particular every Jordan arc extends to a Jordan curve, and no arc separates the plane.*

(3) (Carathéodory extension) *Every conformal homeomorphism $\phi\colon U \to V$ of Jordan disks extends to a homeomorphism $\overline{U} \to \overline{V}$ of their closures.*

(4) (Glued sphere extension) *Every orientation-preserving conformal homeomorphism $\phi\colon U \to V$ of Jordan disks $U, V \subset \mathbb{C}$ extends to an orientation-preserving homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$, and Φ may be chosen with $\Phi(\infty) = \infty$.*

(5) (Fill-equivariance) *Every homeomorphism Φ of $\widehat{\mathbb{C}}$ with $\Phi(\infty) = \infty$ satisfies $\Phi(\mathrm{Fill}(K)) = \mathrm{Fill}(\Phi(K))$ for every compact $K \subset \mathbb{C}$.*

Proof. Clauses (1) and (2) are classical plane topology; see [13, 17]. Clause (3) is Carathéodory's theorem on boundary extension of Riemann maps of Jordan domains; see [14].

For (4), extend ϕ to a homeomorphism $\overline{\phi}\colon \overline{U} \to \overline{V}$ by (3). Its boundary restriction is a homeomorphism $\partial U \to \partial V$ of Jordan curves; the complementary sets $\widehat{\mathbb{C}} \setminus U$ and $\widehat{\mathbb{C}} \setminus V$ are closed Jordan disks by (1), and a homeomorphism between the boundary circles of two closed disks extends over the disks (cone the homeomorphism off an interior point, after identifying each closed disk with the standard one by (1)). Gluing the two extensions along the boundary curves yields a homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ with $\Phi|_U = \phi$. A sphere homeomorphism that preserves orientation on a nonempty open set preserves it globally, so Φ is orientation preserving. Finally, ∞ lies in the interior of the complementary disk $\widehat{\mathbb{C}} \setminus U$, and $\Phi(\infty)$ in the interior of $\widehat{\mathbb{C}} \setminus V$; composing Φ with a homeomorphism of the closed disk $\widehat{\mathbb{C}} \setminus V$ that fixes ∂V pointwise and carries $\Phi(\infty)$ to ∞ (such a homeomorphism exists, interiors of disks being homogeneous relative to the boundary), extended by the identity on $\overline{V}$, arranges $\Phi(\infty) = \infty$ without disturbing $\Phi|_U = \phi$.

For (5), the restriction $\Phi|_{\mathbb{C}}$ is a homeomorphism of $\mathbb{C}$ onto itself. It maps $\mathbb{C} \setminus K$ homeomorphically onto $\mathbb{C} \setminus \Phi(K)$ and permutes complementary components. A homeomorphism of $\mathbb{C}$ is proper, so the image of the unbounded component of $\mathbb{C} \setminus K$ is unbounded and the images of bounded components are bounded; hence $\Phi(\Omega_\infty(K)) = \Omega_\infty(\Phi(K))$, and taking complements as in Lemma 2.1(1) gives $\Phi(\mathrm{Fill}(K)) = \mathrm{Fill}(\Phi(K))$. □

Clause (4) is the mechanism behind every naturality statement in the paper: a univalent change of Jordan localization extends to a sphere homeomorphism, and a sphere homeomorphism preserves all the notions the certificates are built from — compactness, connectedness, fullness (by clause (5) and Lemma 2.1), crosscuts, selected sides, null chains, intersections and disjointness, and, when orientation is preserved, the cyclic order of prime ends and the ribbon structure of an embedded tree. Clause (2) is used in the slit-straightening steps of the repaired slit lemma (Lemma 7.1).

This completes the classical inventory. The dynamical vocabulary of the source manuscript — frozen inventory, free domain, ports, packets, audits — is introduced where it is first needed: the frozen free domain Ω_{free} in Section 7, packets and their audit in Sections 8 and 9.

3. Obstructions to naive carriers

Before any positive theory is developed, it is worth recording exactly why the two most tempting shortcuts fail. The first shortcut is a *selection* rule: given the finite package of marked continua that a carrier must contain, simply take the least compact connected full set containing it. The second is a *completion* rule: given a proposed carrier and the excess produced by pulling it back through the dynamics, simply absorb that excess monotonically until the process stabilizes. Both rules are natural, both are implicit in informal uses of connected hulls, and both fail — the first because the minimizer does not exist, the second because absorption destroys precisely the access data that a protected carrier is supposed to preserve. The two obstructions below are taken from the source manuscript [12, Theorem 6.12, Proposition 6.15]; they are elementary, but they dictate the architecture of everything that follows, and in particular the certificate formalism of Section 5. The comparison with standard continuum-hull constructions in Section 4 should be read with both examples in mind.

3.1. **Failure of least carriers.** Recall from Section 2 that a compact set $K \subset \mathbb{C}$ is *full* if $\mathbb{C} \setminus K$ is connected, and that a *carrier* for a finite package $S \subset \mathbb{C}$ is a compact connected full continuum containing S. Order the carriers of a fixed package by inclusion. A *least* carrier would be a carrier contained in every other carrier of the same package. The following theorem shows that a least carrier fails to exist already for the simplest nontrivial package: two points, with the identity map as dynamics and with no obstacle set whatsoever.

Theorem 3.1 (No least connected full carrier; manuscript Theorem 6.12)**.** *There is no least compact connected full plane continuum containing the two-point package $S = \{-1, 1\}$, even with identity dynamics and no obstacles.*

Proof. Let

$$A_+ = \{e^{i\theta} : 0 \le \theta \le \pi\}, \qquad A_- = \{e^{i\theta} : -\pi \le \theta \le 0\}$$

be the upper and lower semicircles of the unit circle joining -1 to 1. Each is an arc, that is, a homeomorphic image of $[0, 1]$; hence each is compact and connected. An arc does not separate the plane [13], so $\mathbb{C} \setminus A_\pm$ is connected and each $A_\pm$ is full. Each contains S, so each is a carrier for S; in particular the family of carriers is nonempty. Finally, the semicircles meet exactly in their common endpoints:

$$A_+ \cap A_- = S.$$

Suppose a least carrier H existed. By definition of leastness, H is contained in every carrier of S; applying this to the two carriers A_+ and A_- gives

$$H \subset A_+ \cap A_- = S.$$

On the other hand $S \subset H$, since H is a carrier of the package S. Hence $H = S = \{-1, 1\}$. But the two-point set S is disconnected, while a carrier is by definition connected. This contradiction shows that no least carrier exists. □

The failure is one of *leastness*, not of minimality: minimal carriers exist in abundance. Indeed, a subcontinuum of an arc that contains both endpoints is the whole arc, so every arc from -1 to 1 — in particular each of A_+ and A_-, and every circular arc through -1 and 1 — is a minimal element of the family of carriers. The order-theoretic structure is thus the worst possible for a canonical selection: a large family of pairwise distinct minimal elements and no least one.

The moral drawn in the manuscript is the one that governs this paper: "take the minimal connected full hull" is not a carrier rule. Even for two points and trivial dynamics, the carrier is underdetermined, and the ambiguity is not a set-theoretic accident but a genuine topological degree of freedom — the two semicircles pass on opposite sides of the origin, and in any configuration with obstacles, additional atoms, or protected accesses, that choice of side is visible to the rest of the construction. A regulated connector system, together with its access or isotopy class and a forward-coherence proof, is therefore *additional data*, beyond anything determined by the package itself. Recording that data explicitly, and saying precisely when it can be realized by an actual carrier, is the task of the certificate formalism of Section 5.

3.2. **An analytic protected-access collision.** Theorem 3.1 rules out selecting the carrier by minimization. The remaining naive rule is completion:

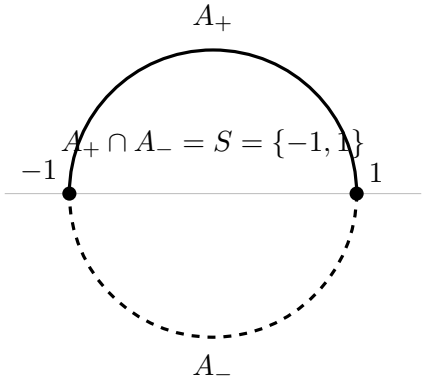


FIGURE 1. Two minimal carriers for the package $S = \{-1, 1\}$: the upper semicircle A_+ (solid) and the lower semicircle A_- (dashed). Each is an arc, hence a compact connected full continuum containing S, and they meet only in the package points. A least carrier would be contained in their intersection S, which is disconnected; hence no least carrier exists (Theorem 3.1).

accept an arbitrary proposed carrier and then absorb, monotonically, whatever excess the dynamics pulls back onto it. The manuscript's Section 6.4 shows that this rule fails as well, and fails in the most favorable analytic circumstances imaginable: a single conformal, critical-value-free, degree-one branch of the quadratic map, acting on an interval. The absorption step erases a protected prime-end access, and monotonicity then propagates the erasure forever.

We reproduce the analytic setup exactly. Let

$$f(z) = z^2, \qquad I = [1, 2], \qquad m = \tfrac{3}{2},$$

and let

$$Q = \Big\{ z : \big|z - \tfrac{3}{2}\big| \le \tfrac{1}{2},\ \operatorname{Im} z \ge 0 \Big\} \tag{4}$$

be the closed upper half-disk of radius $\frac{1}{2}$ centered at m. Set

$$U^- = \big\{z : \big|z - \tfrac{3}{2}\big| < 1\big\}, \qquad U^+ = f(U^-).$$

The disk U^- lies in the half-plane $\operatorname{Re} z > \frac{1}{2}$. Two consequences follow. First, $0 \notin U^-$, so $f'(z) = 2z \neq 0$ on U^- and the critical point of f is outside the domain; since $f(z) = f(w)$ forces $w = \pm z$, and U^- is disjoint from its antipode $-U^- \subset \{\operatorname{Re} z < -\frac{1}{2}\}$, the restriction

$$F = f|_{U^-} : U^- \longrightarrow U^+ \tag{5}$$

is injective, hence a conformal homeomorphism onto U^+. Second, the critical value $0 = f(0)$ is not in U^+, so the branch is critical-value-free. Take

$$T_0 = I, \qquad T_0' = Q, \qquad T_1 = f(I) = [1, 4], \qquad T_1' = f(Q). \tag{6}$$

Within this branch the stage-0 data are the pullbacks of the stage-1 data:

$$T_0' = F^{-1}(T_1'), \qquad T_0 = F^{-1}(T_1). \tag{7}$$

Thus $T_0 = I$ is the proposed carrier at stage 0, and $T_0' = Q$ is the actual pulled carrier: the completion rule under scrutiny would replace T_0 by an enlargement absorbing the difference. Two pieces of terminology from the manuscript's completion formalism are needed, and we define them at first use. The *pulled excess* at stage 0 is the difference $X_0 = T_0' \setminus T_0$ between the pulled carrier and the proposed carrier. The *attachment trace* of the excess on the proposed carrier is

$$\tau_{T_0}(X_0) := \overline{X_0}^{T_0'} \cap T_0, \tag{8}$$

where $\overline{X_0}^{T_0'}$ denotes the closure of X_0 taken in the actual pulled carrier T_0': the trace records where the excess attaches to the proposed carrier after closure, not where it meets it as a set.

Proposition 3.2 (Quadratic protected-access collision; manuscript Proposition 6.15)**.** *In the setup* (4)–(7)*, the pulled excess has the single component*

$$X_0 = Q \setminus I$$

with attachment trace

$$\tau_{T_0}(X_0) = \overline{X_0}^{T_0'} \cap T_0 = I.$$

The closure is taken in the actual pulled carrier T_0'; in particular, the trace is not $X_0 \cap T_0$, which is empty by the definition of the excess. The complete one-stage itinerary of the excess is

$$X_0 \longmapsto f(X_0),$$

through the conformal right-hand branch F, with no critical encounter. Let $p_0^+(m)$ denote the upper landing access to I at m, represented by

$$a_0(t) = m + it, \qquad 0 < t \le \tfrac{1}{4}.$$

Every compact enlargement $H_0 \Subset U^-$ with $I \cup X_0 \subset H_0$ erases this access. Its image enlargement $F(H_0)$ likewise erases the transported access $F_* p_0^+(m)$, by which we mean the landing access represented by $F \circ a_0$.

Proof. The excess and its trace. A point of Q lies in $I = [1, 2]$ exactly when it is real, so

$$X_0 = Q \setminus I = \left\{ z : \left| z - \tfrac{3}{2} \right| \le \tfrac{1}{2},\ \operatorname{Im} z > 0 \right\}$$

is the relative upper half-disk with its real diameter removed. It is the intersection of the closed disk of radius $\frac{1}{2}$ about m with the open upper half-plane, hence convex, hence connected: the excess has a single component. Its closure in $\mathbb{C}$ is the entire closed half-disk Q: every $x \in (1, 2)$ is the limit of $x + it \in X_0$ as $t \downarrow 0$, while the endpoints are approached, for $0 < \varepsilon \le \frac{1}{2}$,

by $1+\varepsilon+i\varepsilon$ and $2-\varepsilon+i\varepsilon$, respectively. As $Q = T_0'$, the closure of X_0 in T_0' is all of Q, and therefore

$$\tau_{T_0}(X_0) = \overline{X_0}^{T_0'} \cap T_0 = Q \cap I = I :$$

the excess attaches to the proposed carrier along the entire interval, even though $X_0 \cap T_0 = \varnothing$ by construction. This is the reason the completion formalism must work with attachment traces rather than intersections: the naive intersection sees nothing, while the closure sees an attachment along all of I.

The itinerary. Since $0 \notin U^-$, we have $F'(z) = 2z \neq 0$ throughout U^-, and the selected branch avoids the critical value $0 \notin U^+$. The itinerary of the excess therefore consists of the single right-sheet, degree-one step $X_0 \mapsto f(X_0)$ with no critical encounter. There is no hidden multivalence, folding, or critical intrusion anywhere in the example.

Erasure of the protected access. The access $p_0^+(m)$ is the access to I at the interior point m from the upper side, represented by a_0, which lies in $U^- \setminus I$ and lands at m. Let $H_0 \Subset U^-$ be compact and contain $X_0 \cup I = Q$. The set Q contains an upper relative half-neighborhood of m: the set

$$\left\{ z : \left|z - \tfrac{3}{2}\right| < \tfrac{1}{2},\ \operatorname{Im} z \geq 0 \right\} \subset Q$$

is a neighborhood of m in the closed upper half-plane. Consequently, any curve in U^- landing at m from the upper side—that is, eventually in the upper component of a sufficiently small disk about m with I removed—eventually lies in this half-neighborhood, hence in $Q \subset H_0$. No curve in the complement $U^- \setminus H_0$ can therefore land at m from the upper side: the access $p_0^+(m)$ no longer exists as an access from the complement of H_0. Any further enlargement still contains Q and cannot restore that access.

Transport to stage 1. The branch F of (5) is a conformal homeomorphism of U^- onto U^+ carrying I to $T_1 = [1,4]$, Q to T_1', X_0 to the stage-1 excess $T_1' \setminus T_1$, and the representing curve $m+it$ to a curve in $U^+ \setminus T_1$ landing at $f(m) = \frac{9}{4}$ from the F-selected side. The compact image $F(H_0)$ contains $F(Q) = T_1'$. A homeomorphism of the localization carries complement curves to complement curves, so the same argument applied to these image data erases the transported access $F_* p_0^+(m)$ at stage 1. Conformality thus transports the identical conclusion to stage 1, as claimed. □

3.3. **What the obstructions force.** The two obstructions are complementary and together close off every naive route. Theorem 3.1 shows that the carrier cannot be *selected* canonically: there is no least connected full hull, even for two points with identity dynamics. Minimal carriers may exist, as the two semicircles in the proof show, but they need not be unique and neither is least; thus "the minimal carrier" does not specify an object. Proposition 3.2 shows that the carrier cannot be *completed* blindly either: the very first monotone absorption of a pulled excess can erase a protected

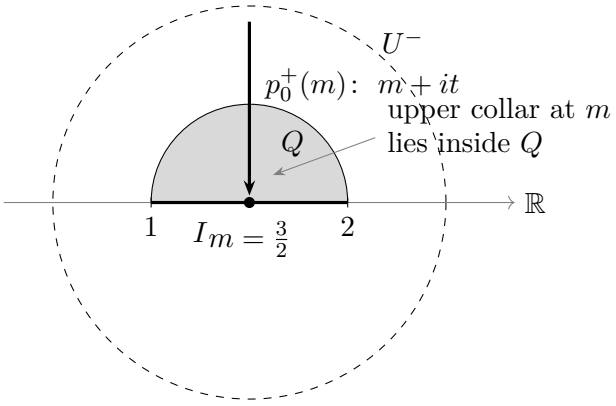


FIGURE 2. The quadratic protected-access collision (Proposition 3.2). The proposed carrier is the interval $I = [1, 2]$ (thick); the pulled carrier is the closed upper half-disk Q (shaded) of radius $\frac{1}{2}$ about $m = \frac{3}{2}$, inside the localization disk U^- (dashed) on which $f(z) = z^2$ restricts to a conformal homeomorphism. The protected upper access $p_0^+(m)$, represented by the vertical segment $m + it$, must traverse Q: absorbing the excess $X_0 = Q \setminus I$ forces the carrier to contain the entire upper collar at m and erases the access.

prime-end access, and monotonicity then makes the loss irreversible at every later stage of the enlargement, as well as at the conformally transported stage.

The second obstruction is the sharper of the two, because of what it does *not* involve. In the manuscript's collision bookkeeping [12], which records sheet mergers, critical intrusions, obstacle collisions, access identifications, and localization escapes in the order $(\chi_{\mathrm{sheet}}, \chi_{\mathrm{crit}}, \chi_{\mathrm{obs}}, \chi_{\mathrm{access}}, \chi_{\mathrm{loc}})$, the collision tuple of the completed enlargement in this example is

$$\big(\varnothing,\ \varnothing,\ \varnothing,\ \{p_0^+(m),\, F_* p_0^+(m)\},\ \varnothing\big) :$$

every coordinate vanishes except the access coordinate. The branch is conformal, degree one, and critical-value-free; there is no sheet merger, no obstacle, and no localization escape. Consequently no amount of critical-point bookkeeping—however carefully one tracks critical points, critical values, or degrees—can detect, let alone excuse, the collision. Access protection is invisible to critical data and must be carried as explicit data of its own, with its own survival theorem.

This is exactly what the remainder of the paper provides. The certificate formalism of Section 5 records the protected accesses as prime-end crosscut chains [14] with pairwise disjoint relative terminal collar tails, together with free-port arcs carrying singleton attachment traces and a ribbon tree with declared sides; Theorem 5.4 then realizes any such certificate by an actual

compact connected full carrier in which every protected chain survives with unchanged cyclic order. The selection ambiguity of Theorem 3.1 is resolved not by minimization but by the certificate itself: the connector system, its access class, and its routing are supplied as data, and the exact free-component criterion of Section 7 determines precisely when such data can be supplied at all.

4. Comparison with standard continuum-hull constructions

The certificate formalism developed from Section 5 onward is offered as a replacement for a family of standard constructions, and it is fair to ask what, exactly, is being replaced and why. This section compares the framework with five natural comparison points: least or minimal connected full hulls; monotone absorption of pulled excess; regulated and Hubbard trees in uniformized or locally connected coordinates; Moore decompositions and monotone quotients; and the lamination-lift candidate of the source manuscript. The two obstructions of Section 3 already dispose of the first two; the remaining three fail for structural reasons that are worth stating precisely, because they explain several design decisions of the certificate formalism that would otherwise look pedantic. Throughout, the comparison is with respect to the task fixed in the introduction: join finitely many pairwise disjoint full atoms inside a Jordan localization, avoid a frozen forbidden set, and *preserve designated prime-end accesses together with their cyclic order*. Constructions that solve a different problem well are not thereby criticized; the claim is only that none of the constructions, in the forms reviewed here, records all of these requirements simultaneously, and that the last of them — the lamination lift — is an open candidate, not a construction.

4.1. **Least and minimal connected full hulls.** The most tempting rule is extremal: "take the least connected full hull of the required package." Theorem 3.1 (manuscript Theorem 6.12) shows the rule names no object, already for the two-point package $S = \{-1, 1\}$ with identity dynamics and no obstacles: the upper and lower semicircles are carriers meeting exactly in S, so a least carrier would be contained in their disconnected intersection. The failure is specific to *leastness*: minimal carriers exist in abundance — every arc from -1 to 1 is one — and form a large antichain. An extremal selection principle therefore cannot even break the tie between the two semicircles, and the tie is not innocent: the two choices pass on opposite sides of the origin, and in any configuration with obstacles or relative protected collar tails that choice of side is visible to every later routing step. The source manuscript sharpens the point dynamically [12, Proposition 6.11]: a package, full connected carriers, critical-value-free degree-one selected-carrier dynamics, and identical coarse orbit labels are realized simultaneously by an interval and by a triod, and attachment-freeness holds in only one of the two models, so no data of the listed kinds can select between them.

The connected-hull descriptions in the renormalization literature are not at fault here, and the comparison should not be read as a criticism of them. Dudko's roadmap describes virtual Julia sets as connected hulls of Cantor families of small Julia sets carrying a finite part of the critical orbit [4]; that description specifies *which* continua a carrier must contain, and it does not purport to specify *how* they are joined. Theorem 3.1 shows that the routing is genuinely additional data — in the manuscript's words, a regulated connector system, together with its access or isotopy class and a forward-coherence proof, is additional data [12] — and the certificate of Definition 5.1 is exactly a format for that data, with Theorem 7.3 deciding when the format can be filled in at all.

4.2. **Monotone absorption of pulled excess.** The second rule is completion: accept any proposed carrier and absorb, monotonically, whatever excess the dynamics pulls back onto it, until the process stabilizes. Unlike the least hull, this rule is well defined: the manuscript proves that the finite inflationary completion operator stabilizes at a least fixed point, unique and conjugation-equivariant [12, Theorem 6.14]. The defect is not existence but what completion destroys. Proposition 3.2 (manuscript Proposition 6.15) exhibits a conformal, critical-value-free, degree-one branch of $z \mapsto z^2$ in which the very first absorption of pulled excess — the relative upper half-disk attaching to the interval along its entire length — erases a protected prime-end access and its conformally transported copy. Monotonicity, the rule's defining virtue, is precisely what makes the loss permanent: enlargement can only add points, and no addition restores a complement access through a set already absorbed. Moreover, every coordinate of the manuscript's collision bookkeeping except the access coordinate vanishes in this example, so no amount of critical-value, degree, or sheet bookkeeping detects the erasure (Section 3.3).

The certificate formalism keeps monotone completion but reverses the order of commitments. The relative protected collar tails are frozen into the inventory Q *before* any absorption is contemplated; a proposed absorption that would enter such a tail is not a step of a completion but a named violation, and the persistence clause of Theorem 7.3, together with the audit of Theorem 9.3, converts each such violation into a certificate — a merger, a port loss, a separation — that survives every further monotone enlargement. Absorption remains available for everything unprotected; what changes is that "protected" is a constraint checked by the plane, not an intention held by the author.

4.3. **Regulated trees in uniformized coordinates and Hubbard trees.** For a postcritically finite polynomial P with connected filled Julia set, the classical Hubbard tree is the regulated hull in $K(P)$ of the finite union of the forward critical orbits [3]. Arbitrary finite invariant sets, or local connectivity alone, do not define the classical Hubbard tree. The construction uses regulated arcs, uniformized coordinates, and canonical

internal arcs. Local connectivity gives a continuous extension of the exterior Riemann map to the boundary; that extension is a homeomorphism only when the boundary is a Jordan curve [3, 14]. Kahn–Lyubich impose finite-decoration combinatorics and prove a priori bounds for that class, using puzzle pieces, rays, and equipotentials in the construction [8].

None of these steps is guaranteed by the standing hypotheses of the present paper. The atoms are arbitrary full continua; no Riemann map is asserted to extend, no external or internal ray is asserted to land, and regulated arcs need not exist. This is not a hypothetical worry but the operating regime of the intended applications, in which local connectivity of the relevant continua is exactly what is not known. The closest publicly available construction is instructive as a boundary marker. In its stated primitive-renormalization setting, the preprint of Kahn–Kapiamba–Lyubich uses a rational-lamination quotient, a locally connected combinatorial model, and the associated natural projection [7, §§2.5–2.6]. This is not a general construction for arbitrary full continua. As the source manuscript's own boundary records [12], the preprint is a positive control in its primitive-renormalization category, not a theorem for the arbitrary packages contemplated here. The framework of this paper is calibrated to the complementary regime: Theorem 5.4(6) uses no local connectivity of any atom, the tree of the certificate is produced by the exact criterion of Theorem 7.3 from planar data alone, and no canonicity of the spanning tree is claimed anywhere — the tree is data, chosen in a free component whose existence is the actual content. What is lost relative to the Hubbard-tree regime is canonicity; what is gained is existence without a locally connected model whenever the exact free-component criterion holds.

4.4. **Moore decompositions and monotone quotients.** A third classical route collapses instead of routing. Moore's theorem states that if $\mathcal{G}$ is an upper-semicontinuous decomposition of S^2 into nonseparating continua and the quotient has more than one point, then $S^2/\mathcal{G}$ is homeomorphic to S^2 [10]. One is therefore tempted to collapse every atom (or every completed packet) to a point, join the finitely many image points by arcs in the quotient sphere — where the joining problem is trivial — and pull the picture back. For the connection problem *without* protected data this can be a serviceable device and is one standard mechanism for passing to a simpler quotient model.

For the access-preserving problem the unmarked quotient is not, by itself, a certificate. Collapsing an atom sends all of its points to one quotient point, and neither that point nor a tree incident to it records which named prime ends, selected sides, or relative collar tails are to be protected. Crosscuts in the complement can have quotient images, and in favorable decompositions distinct approaches may still be recoverable; the point is that Moore's theorem alone supplies none of the required marking or survival conclusions. Those data would have to be transported separately.

Likewise, the preimage of an arc under a continuous monotone quotient is a continuum, but generally a saturated one. It does not automatically have singleton attachment traces, retain selected prime-end accesses, or preserve marking data; a fiber can contain more attachment material or meet a forbidden collar. Thus a Moore quotient may support a construction only after it is augmented by the same access and incidence data that the certificate formalism records explicitly. The present realization works in the original plane, carries protected accesses as crosscut chains with selected sides, and attaches its connector at declared free ports with singleton traces (Theorem 5.4(2)).

4.5. **The lamination-lift candidate.** The most serious competitor is the manuscript's own candidate, recorded in its Section 10.3 [12], and the comparison must be exact: the candidate is presented there, and here, as a candidate only. Suppose, following the non-locally-connected construction of Kahn–Kapiamba–Lyubich [7], that an ambient carrier admits a monotone semiconjugacy

$$\pi\colon K \longrightarrow K_{\mathrm{com}}$$

onto a locally connected combinatorial model, together with a forward-coherent regulated tree $T_{\mathrm{com}} \subset K_{\mathrm{com}}$ containing the prescribed finite marked set. The canonical candidate connector is

$$H_{\mathrm{lam}} \;=\; \mathrm{Fill}\big(\pi^{-1}(T_{\mathrm{com}})\big), \tag{9}$$

equation (9.3) of [12]. Part of what is wanted is genuinely true, by soft arguments: a monotone preimage of a compact connected set is compact and connected, so (9) is a full continuum containing the preimage of the prescribed marked set; the semiconjugacy provides forward coherence before fullification; and fullification preserves genuine polynomial forward invariance.

The missing assertion is not connectedness. Measured against the certificate clauses, two failure channels are named in the manuscript, and both remain unresolved. First, a projection fiber $\pi^{-1}(x)$ over a point of the tree, or a component added by fullification, may enter a protected collar otherwise than at a declared free singleton port; this is the erasure mechanism of Proposition 3.2 wearing different clothes, and it is instructive that the manuscript rejects the upper-half-disk collision at exactly this clause, the excess occupying the protected upper collar with an interval attachment trace [12]. Second, a disconnected stage package may still split across iterated pullback components, so that containing the prescribed marked set does not yet realize the complete package routing. The manuscript collects the clauses a proof would have to establish — authorized localization and complete frozen inventory, collar avoidance except at declared free singleton ports, realization of every package component and route, obstacle handling inside the localization, complete access marking, and an incidence bound or

compression — as its Open Problem 10.6 (Virtual Protected Routing Certificate). That problem is open, and nothing in this paper resolves any part of it; consistently with the manuscript's Warning 10.59, the candidate (9) is discussed here only to locate it on the map.

What this paper contributes to the candidate is the deciding instrument, not a verdict. For any frozen inventory that a future argument supplies, Theorem 7.3 decides certificate existence exactly; Theorems 8.4 and 9.3 and Proposition 9.5 make the decision checkable packet by packet; and the two failure channels above fall under the typed, monotone-persistent defect certificates of Section 9 — a fiber in a collar is a port-loss or merger defect, a split package is a separation in the sense of the persistence clause of Theorem 7.3. Whether the candidate passes the test is precisely what is open.

4.6. **Summary.** Table 1 compresses the comparison. The pattern is uniform: each standard construction either fails to exist (the least hull), can destroy the protected data (absorption), does not by itself record those data (Moore quotients), requires the hypothesis whose absence is the whole difficulty (regulated and Hubbard trees), or is an explicitly open candidate (the lamination lift). The certificate framework renounces canonicity — the connector is data, not an invariant — and in exchange obtains, for arbitrary full atoms and with no local connectivity of the atoms: realization of every supplied certificate with access survival and cyclic-order preservation (Theorem 5.4); an exact existence criterion on frozen data (Theorem 7.3); checkable sufficient tests (Theorem 8.4, Proposition 9.5); order-independent completion with a three-clause audit (Theorem 9.3); and failure certificates that persist under monotone enlargement. The effect is to convert "choose a hull," an instruction that Theorem 3.1 shows to be ill posed and Proposition 3.2 shows to be destructive, into a checkable planar test with a stable answer in both directions. This is the precise sense in which the contribution of the paper is proof technology: no new theorem of classical planar topology is claimed, and every statement about the open constructions touched on above — virtual carriers, lamination lifts, and their dynamical certificates — remains exactly as open here as in the source manuscript.

Construction	Existence	Access guarantee	Failure witness
least connected full hull	no (Thm. 3.1)	—	—
monotone absorption	least fixed point [12]	erasable (Prop. 3.2)	post hoc only
regulated/Hubbard tree	needs locally connected model [2, 7]	native category only	—
Moore monotone quotient	specified nontrivial u.s.c. decomposition into nonseparating continua [10]	markings not automatic	—
lamination lift (9)	candidate [12]	open	open
protected certificate	exact criterion (Thm. 7.3)	preserved (Thm. 5.4)	persistent (Thm. 9.3)

TABLE 1. The five standard constructions against the certificate framework, measured on the access-preserving connection task. "Existence" asks whether the construction produces a well-defined carrier for arbitrary pairwise disjoint full atoms in a Jordan localization; "access guarantee" asks whether designated prime-end data survive with their cyclic order; "failure witness" asks whether a negative outcome is witnessed by a certificate that persists under monotone enlargement. The lamination-lift row records candidacy, not failure: its clauses are the manuscript's open problem, and no verdict is implied in either direction.

5. Protected ribbon-routing certificates and realization

The two obstructions of the preceding section calibrate what a realization theorem can promise. Theorem 3.1 shows that even the two-point package with identity dynamics and no obstacles admits no least compact connected full carrier, so no canonical minimal choice is available and a connector must be *supplied*, not selected by an extremal principle. Proposition 3.2 shows that a compact enlargement absorbing the pulled excess can permanently erase a protected prime-end access even along a conformal, critical-value-free branch: once a set occupying a protected side has been absorbed, no further monotone enlargement restores an access through it. Together these force the shape of the present section. The connector data must be recorded *in advance*, as an explicit finite certificate that names every protected access, reserves a collar around each one, and routes the connecting tree through the reserved-free region. The theorem then owed to the reader is purely topological: every such certificate is realized by a compact connected full

carrier, the protected accesses survive as prime ends of the enlarged complement, distinct accesses stay distinct, their cyclic order is unchanged, and the two regular-neighborhood side-readings of each connector edge appear in the boundary order dictated by the ribbon structure of the tree. This is the content of Theorem 5.4 below. The section closes with the naturality of the construction under univalent changes of the Jordan localization (Proposition 5.6). The question of when a certificate *exists* is deferred to Section 7, where it receives an exact answer.

Setup. Throughout this section $U \subset \mathbb{C}$ is a Jordan disk, that is, a bounded simply connected domain whose boundary is a Jordan curve. A *continuum* is a nonempty compact connected subset of $\mathbb{C}$; a continuum K is *full* if $\mathbb{C} \setminus K$ is connected, equivalently if $\widehat{\mathbb{C}} \setminus K$ is a simply connected domain containing ∞. Following the source manuscript [12], the full continua to be joined are called *atoms*. Fix $s \geq 1$, and let

$$A_1, \dots, A_s \Subset U$$

be pairwise disjoint atoms, and let

$$Z \Subset U \setminus \bigcup_i A_i$$

be the compact frozen forbidden set. It bundles every non-collar obstacle, critical-value exclusion, and localization exclusion. Nothing in this section uses any dynamical structure on the A_i or on Z; the data are purely planar.

For each nondegenerate atom A_i, the domain $\Omega_i = \widehat{\mathbb{C}} \setminus A_i$ is simply connected and the prime-end formalism of Section 2.4 applies [14]. Protected chains are permitted only at nondegenerate atoms; a degenerate atom may still carry a free port and participate in the carrier, but its protected-label set is empty. All chains below are *null chains* in the sense of (2):

$$\overline{D_{n+1}} \setminus A_i \subset D_n, \qquad \operatorname{diam}_{\mathrm{sph}}(\gamma_n) \longrightarrow 0.$$

The record of a chain contains, for every n, the declaration of which side D_n of γ_n is selected. It is exactly this declaration that the certificates below protect.

For every protected access to a nondegenerate A_i, indexed by a finite set of labels a specifying pairwise distinct prime ends at that fixed atom, choose a prime-end crosscut chain

$$\Gamma_{i,a} = (\gamma_{i,a,n},\, D_{i,a,n})_{n \geq 1}$$

of $\widehat{\mathbb{C}} \setminus A_i$. The chains are truncated, that is, an initial segment of each is discarded and the indices shifted, so that the terminal *relative collar tails*

$$\overline{D_{i,a,1}} \setminus A_i$$

are pairwise disjoint, lie in U, and miss Z and the other atoms. For each fixed atom, Hausdorff separation in its prime-end closed disk gives pairwise disjoint sufficiently deep prime-end neighborhoods. Across different atoms,

choose pairwise disjoint Euclidean neighborhoods $V_i \Subset U$ of the compact sets A_i, all disjoint from Z. Because the nested compact sets $\overline{D_{i,a,n}}$ decrease to their impressions in A_i, sufficiently deep members lie in V_i. A common finite truncation therefore has all the stated properties. Their closures are allowed to meet inside a common atom, so overlapping prime-end impressions are not excluded. The set $D_{i,a,1}$ is called the (terminal, or first) *protected collar* of the access (i, a); every deeper selected domain $D_{i,a,n}$ lies inside it. Figure 3 shows a single atom, one protected chain, its nested selected domains, and its collar.

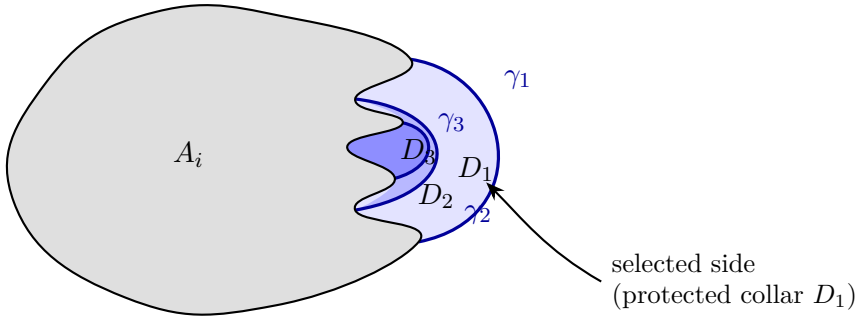


FIGURE 3. A protected prime-end crosscut chain at a single atom A_i. The crosscuts $\gamma_1, \gamma_2, \gamma_3$ are pairwise disjoint arcs of $\widehat{\mathbb{C}} \setminus A_i$ with endpoints on ∂A_i and shrinking diameters; the shaded regions are the nested selected sides $D_1 \supset D_2 \supset D_3$, drawn with increasing opacity toward the prime end. The outermost selected domain D_1 is the protected collar of the access: after truncation the relative collar tail $\overline{D_1} \setminus A_i$ is disjoint from every other relative tail, lies in U, and misses the obstacle set Z and the other atoms. The boundary of the atom is drawn deliberately oscillating: no local connectivity of ∂A_i is assumed anywhere in this section, and the access is carried entirely by the chain, not by any landing curve.

Definition 5.1 (Protected ribbon-routing certificate; manuscript Definition 10.1)**.** A *protected ribbon-routing certificate* consists of the preceding data—the Jordan disk U, the atoms $A_1, \dots, A_s$, the forbidden set Z, and the truncated protected chains $\Gamma_{i,a}$—together with pairwise disjoint Jordan embeddings $\alpha_i \colon [0,1] \hookrightarrow U$, called *free-port arcs*, satisfying

$$\alpha_i \cap A_i = \{x_i\}, \qquad \alpha_i((0,1]) \cap \Big(Z \cup \bigcup_j A_j \cup \bigcup_{j,a} (\overline{D_{j,a,1}} \setminus A_j) \Big) = \varnothing,$$

where $x_i = \alpha_i(0)$, and an embedded finite tree $G \subset U$ joining the outer port endpoints $\alpha_1(1), \dots, \alpha_s(1)$ while missing the same protected and forbidden

sets $Z \cup \bigcup_j A_j \cup \bigcup_{j,a} (\overline{D_{j,a,1}} \setminus A_j)$, subject to

$$G \cap \alpha_i = \{\alpha_i(1)\} \quad (1 \le i \le s), \qquad G \cup \bigcup_i \alpha_i \text{ an embedded finite plane tree.}$$

As usual, a finite tree may be degenerate, consisting of one vertex and no edges; this convention is used when $s = 1$. The planar cyclic order of half-edges at every vertex of G is retained as part of the certificate (its *ribbon structure*). Every access read by a proof must be a protected chain $\Gamma_{i,a}$; every connector-side reading must be a declared side of this ribbon tree, interpreted through the boundary of a thin regular neighborhood.

Given a certificate R, define its *carrier*

$$H_R \;=\; G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \tag{10}$$

The finite plane tree

$$T_R := G \cup \bigcup_i \alpha_i$$

is its *connector tree*, and $T_R \cap A_i$ is the *connector attachment trace* at A_i.

Four reading notes. First, the port arc α_i touches its atom at the single *free port* x_i, and its positive-parameter part avoids every protected relative collar tail, including those based at the same atom. The abstract certificate does not require x_i to avoid a protected impression inside A_i; applications that need that stronger marking condition impose it explicitly through the unprotected exposure locus of Section 9. Second, "joining the outer port endpoints" is read as part of the embeddedness of the certificate: the tree G meets each port arc α_i in exactly the point $\alpha_i(1)$, so that $G \cup \bigcup_i \alpha_i$ is again an embedded finite tree, with the port arcs attached as pendant edges. This is the configuration produced by the construction of Section 7, where the routing tree is built in the complement of the slit port arcs, and it is used in the proof of the realization theorem below. Third, the tree G is a combinatorial object with a planar enhancement: recording the cyclic order of half-edges at each vertex fixes, up to isotopy, the two sides of every edge, and it is these *declared ribbon sides* that the realization theorem reads, through thin regular neighborhoods, in a prescribed boundary order. No prime end associated with an individual tree edge is asserted. Fourth, the final sentence of the definition is a discipline on all later arguments, not a property of the data: any subsequent proof that consults an access of the carrier must consult a protected chain, and any proof that consults a connector side must use a declared ribbon side rather than an incidental feature of a particular embedding. Figure 4 shows a certificate with three atoms.

Fullness of one-point tree attachments. The first step of the realization is to verify that the carrier (10) is a full continuum. The union $T = G \cup \bigcup_i \alpha_i$ is itself a finite plane tree: G is a finite tree and, by the attachment

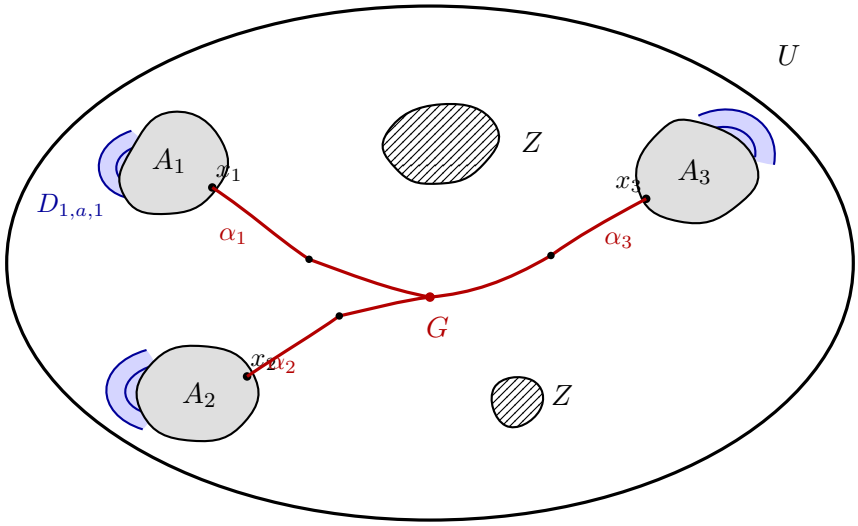


FIGURE 4. A protected ribbon-routing certificate with three atoms inside the Jordan disk U. Each atom A_i carries a protected collar (shaded, with two nested crosscuts indicated) and a free-port arc α_i (heavy) landing at the single free point x_i; its positive-parameter part avoids every relative collar tail. The embedded finite tree G (heavy) joins the outer port endpoints; the positive-parameter parts of the α_i and all of G avoid the hatched obstacle set Z, all atoms, and all relative terminal collar tails. The planar cyclic order of the half-edges at the vertex of G is part of the certificate. The carrier is $H_R = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i$.

convention recorded after Definition 5.1, each α_i meets G in exactly the point $\alpha_i(1)$, so adjoining the arcs one at a time adds pendant edges and creates no cycle. By the port conditions of Definition 5.1, T meets each atom A_i in exactly the point x_i and otherwise misses every atom. The relevant general fact is the following lemma, for which we record two proofs: a direct induction on Janiszewski's separation theorem, and a cohomological argument by Alexander duality and Mayer–Vietoris. The first is elementary and self-contained; the second is the pattern that recurs, for pairwise disjoint packets, in the packet lemmas of Section 8.

Lemma 5.2 (One-point tree attachments are full; manuscript Lemma 10.2)**.** *If a finite plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, then their union is a full continuum.*

First proof (Janiszewski induction). Let T be the finite plane tree and $A_1, \dots, A_s$ the pairwise disjoint full continua, with $T \cap A_i = \{p_i\}$. All sets are regarded as compact subsets of the sphere $\widehat{\mathbb{C}}$, and *nonseparating* means

that the complement in $\widehat{\mathbb{C}}$ is connected; for a compact subset of $\mathbb{C}$ this is equivalent to fullness, since the complement in $\widehat{\mathbb{C}}$ is open and connected exactly when the bounded complementary components in $\mathbb{C}$ are absent.

A finite plane tree is nonseparating. Indeed, T is compact, connected, and simply connected, and one may argue by induction on the number of edges: a zero-edge tree is a singleton and a one-edge tree is a closed arc, neither of which separates the sphere; and if $T = T' \cup e$ with e a closed pendant edge meeting T' in one endpoint, then T' and e are nonseparating compacta with connected (one-point) intersection, so Janiszewski's theorem, quoted below, makes T nonseparating.

We now attach the full continua one at a time. Put

$$B_0 = T, \qquad B_k = T \cup A_1 \cup \cdots \cup A_k \quad (1 \le k \le s),$$

so that B_s is the union in question. Each B_k is compact, and it is connected because T is connected and each attached A_j meets T. We claim every B_k is nonseparating. The case $k = 0$ was just proved. Assume B_{k-1} is nonseparating and consider $B_k = B_{k-1} \cup A_k$. The intersection is

$$B_{k-1} \cap A_k = (T \cap A_k) \cup \bigcup_{j<k} (A_j \cap A_k) = \{p_k\},$$

a singleton, hence connected: the atoms are pairwise disjoint and T meets A_k only at p_k. Both B_{k-1} and A_k are compact and nonseparating. Janiszewski's separation theorem [13, 17] states that if two closed subsets of the sphere have connected intersection and neither separates a given pair of points, then their union does not separate that pair. Applying this to every pair of points of $\widehat{\mathbb{C}} \setminus B_k$ shows that B_k separates no pair of points of its complement, that is, B_k is nonseparating. By induction, B_s is a compact connected set with connected complement in $\widehat{\mathbb{C}}$, hence a full continuum. □

Second proof (Alexander duality and Mayer–Vietoris). Work again in $\widehat{\mathbb{C}} = S^2$ and retain the notation B_k of the first proof. For a compact set $X \subset S^2$, Alexander duality in its Čech form [15, Chapter 6] gives an isomorphism

$$\widetilde{H}_0\big(S^2 \setminus X; \mathbb{Z}\big) \;\cong\; \check{H}^1(X; \mathbb{Z}),$$

so X is nonseparating precisely when its first reduced Čech cohomology vanishes. (Čech cohomology is essential here: the atoms are arbitrary continua and need not be locally connected or of CW homotopy type; Čech theory has the required continuity under inverse limits of neighborhoods and satisfies the Mayer–Vietoris sequence for compact pairs in a normal ambient space.)

The tree T is contractible, so $\check{H}^1(T; \mathbb{Z}) = 0$; and $\check{H}^1(A_i; \mathbb{Z}) = 0$ for every i because each A_i is full, by the same duality read in the other direction. Now induct on k. Suppose $\check{H}^1(B_{k-1}; \mathbb{Z}) = 0$ and write $B_k = B_{k-1} \cup A_k$ with $B_{k-1} \cap A_k = \{p_k\}$. The Mayer–Vietoris sequence in reduced Čech cohomology for the compact pair (B_{k-1}, A_k) reads

$$\widetilde{\check{H}}^{\,0}(\{p_k\}; \mathbb{Z}) \longrightarrow \check{H}^1(B_k; \mathbb{Z}) \longrightarrow \check{H}^1(B_{k-1}; \mathbb{Z}) \oplus \check{H}^1(A_k; \mathbb{Z}).$$

The left group vanishes because the one-point intersection is connected, and the right group vanishes by the inductive hypothesis and fullness of A_k. Exactness forces $\check{H}^1(B_k;\mathbb{Z}) = 0$. By induction $\check{H}^1(B_s;\mathbb{Z}) = 0$, and Alexander duality converts this back into connectedness of $S^2 \setminus B_s$. Since B_s is compact and connected, it is a full continuum. □

The two proofs establish literally the same induction step, and the translation between them is the dictionary used throughout this paper: "nonseparating" equals "vanishing first Čech cohomology," "connected intersection" equals "vanishing zeroth reduced cohomology of the overlap." The cohomological form is the one that survives when the finitely many one-point overlaps are replaced by the pairwise *disjoint* packets of Section 8, where the nerve argument of Lemma 8.3 takes over.

Survival of protected chains. The second step is the heart of the realization: enlarging an atom to a full continuum that stays out of a protected collar does not damage the protected access. The chain survives verbatim—the same crosscuts, the same selected sides—as a prime-end chain of the complement of the enlarged continuum, and finitely many protected accesses with disjoint relative collar tails stay distinct and keep their cyclic order.

Lemma 5.3 (Protected crosscut survival and order; manuscript Lemma 10.3)**.** *Let $H \supset A_i$ be a full continuum and let $(\gamma_n, D_n)_{n\geq 1}$ be one of the protected chains above. If*

$$(H \setminus A_i) \cap (\overline{D_1} \setminus A_i) = \varnothing,$$

then the same crosscuts form a prime-end chain in $\widehat{\mathbb{C}} \setminus H$. Their endpoints remain boundary points and their selected sides are the old domains D_n. A finite family of such chains with pairwise disjoint relative terminal collar tails induces distinct prime ends of $\widehat{\mathbb{C}} \setminus H$ with unchanged cyclic order. The order comparison is made for a family carried by one source complement; no identification of the prime-end circles of different atoms is asserted.

Proof. Since H is a full continuum, $\widehat{\mathbb{C}} \setminus H$ is a simply connected domain and has its own prime-end theory. We verify that the entire chain $(\gamma_n)_{n\geq 1}$, with the same selected sides, is a null chain for it.

The selected domains avoid H entirely. For every n the selected domain D_n lies in $\widehat{\mathbb{C}} \setminus A_i$, so $D_n \subset \overline{D_n} \setminus A_i \subset \overline{D_1} \setminus A_i$ by the null-chain relation. Hence

$$H \cap D_n = (A_i \cap D_n) \cup \big((H \setminus A_i) \cap D_n\big) \subset \varnothing \cup \big((H \setminus A_i) \cap (\overline{D_1} \setminus A_i)\big) = \varnothing,$$

so $D_n \subset \widehat{\mathbb{C}} \setminus H$ for every n.

Each crosscut remains a crosscut. Fix $n \geq 1$. For $n = 1$, $\gamma_1 \subset \overline{D_1} \setminus A_i$; for $n \geq 2$, the null-chain relation gives $\gamma_n \subset \overline{D_n} \setminus A_i \subset D_{n-1} \subset D_1$. The collar-clearance hypothesis therefore makes the open arc γ_n miss H; its two distinct endpoints lie on ∂A_i, hence in $A_i \subset H$. Every neighborhood of either

endpoint meets the crosscut interior, which consists of points of $\widehat{\mathbb{C}} \setminus H$, so both endpoints remain boundary points of the enlarged complement $\widehat{\mathbb{C}} \setminus H$, and they are still distinct. Thus γ_n is again an open Jordan arc in $\widehat{\mathbb{C}} \setminus H$ whose closure is an arc with two distinct endpoints on $\partial(\widehat{\mathbb{C}} \setminus H)$, that is, a crosscut of $\widehat{\mathbb{C}} \setminus H$.

The selected sides are unchanged. For $n \geq 1$ let D'_n denote the component of $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$ containing D_n; this is well defined because D_n is connected and disjoint from $H \cup \gamma_n$. We claim $D'_n = D_n$, so that the containment $D'_n \supset D_n$ asserted in the statement holds with equality. The set D_n is open, and it is relatively closed in $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$: as a component of the open set $(\widehat{\mathbb{C}} \setminus A_i) \setminus \gamma_n$ it satisfies

$$\partial D_n \subset \partial A_i \cup \overline{\gamma_n} \subset A_i \cup \gamma_n \subset H \cup \gamma_n,$$

since the closure of the open arc γ_n adds only its two endpoints, which lie in A_i; hence no boundary point of D_n belongs to $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$, and every point of that set lying in $\overline{D_n}$ already lies in D_n. A nonempty connected subset of an open set that is both open and relatively closed is a component of it, so $D'_n = D_n$. Nestedness of the new sides is now inherited from the original chain: $D'_{n+1} = D_{n+1} \subset D_n = D'_n$ for every $n \geq 1$, and $\gamma_{n+1} \subset D_n = D'_n$ lies in the selected side of γ_n, as a null chain requires.

The diameters still tend to zero. The crosscuts are unchanged as point sets, so their spherical diameters are the same numbers as before and still tend to zero. Moreover the original null-chain relation gives

$$\overline{D_{n+1}} \setminus H \subset \overline{D_{n+1}} \setminus A_i \subset D_n.$$

Consequently $(\gamma_n, D'_n)_{n\geq 1}$ is the same null prime-end chain in $\widehat{\mathbb{C}} \setminus H$, which proves the first two assertions.

Distinctness. Now consider a finite family of protected chains satisfying the hypothesis, with pairwise disjoint relative terminal collar tails. Their terminal selected domains are then pairwise disjoint. Recall that two null chains of the same domain define the same prime end if and only if they eventually divide one another: every selected domain of either chain contains all sufficiently deep selected domains of the other. Suppose two chains of the family, with terminal domains $D_1^{(1)}$ and $D_1^{(2)}$, induced the same prime end of $\widehat{\mathbb{C}} \setminus H$. Then some deep selected domain of the first chain—by the preceding step, the same set $D_m^{(1)}$ as before the enlargement—would be contained in a selected domain of the second chain, hence in $D_1^{(2)}$; but that deep domain is nonempty and contained in $D_1^{(1)}$, contradicting $D_1^{(1)} \cap D_1^{(2)} = \varnothing$. Disjoint terminal domains therefore prevent mutual division, and the induced prime ends are pairwise distinct. In particular the survival map, from protected chains to prime ends of $\widehat{\mathbb{C}} \setminus H$, is injective on such a family.

Cyclic order. The prime ends of the simply connected domain $\widehat{\mathbb{C}} \setminus H$ carry a natural cyclic order, the one transported from the unit circle by a

Riemann map [14]. For a finite family of prime ends represented by pairwise disjoint crosscuts whose selected sides are also pairwise disjoint, this order can be read off inside the domain, as follows. Choose one deep crosscut from each chain, deep enough that the chosen crosscuts lie in the pairwise disjoint terminal collars and, in the prime-end closed disk, the closures of the corresponding cut-off caps are pairwise disjoint. This is possible because the represented prime ends are distinct and that closed disk is Hausdorff. Thus the chosen crosscuts and selected sides are pairwise disjoint. Fix three chains of the family, with chosen crosscuts c_1, c_2, c_3 and selected sides S_1, S_2, S_3. Call an embedded triod Y—three arcs with a common branch point, disjoint except at that point—*computing* if Y lies in the domain and meets the configuration $\bigcup_{k=1}^{3}(c_k \cup S_k)$ in exactly its three leg tips, one tip on each c_k. Transporting by a Riemann map turns the configuration into three pairwise disjoint crosscuts of the unit disk cutting off pairwise disjoint caps, and a computing triod into an embedded triod tipping on the three crosscuts and otherwise avoiding crosscuts and caps. In the disk, computing triods exist, and the cyclic order of the three legs at the branch point of any computing triod agrees with the cyclic order of the three caps on the circle, that is, with the cyclic order of the three prime ends: the complement of the three open caps in the closed disk is a closed Jordan region whose boundary traverses the three crosscuts in the cyclic order of the caps, and an embedded triod in a Jordan region whose legs end on three disjoint boundary arcs realizes at its branch point exactly the cyclic order of those arcs. Hence, back in the original domain, computing triods exist for every triple of chains, and each one realizes the cyclic orientation of its prime-end triple at its branch point.

Now compare the two domains. By the preceding steps, the chosen crosscuts and their selected sides are *literally the same sets* for $\widehat{\mathbb{C}} \setminus A_i$ and for $\widehat{\mathbb{C}} \setminus H$. Fix a triple of chains and choose a computing triod Y in the smaller domain $\widehat{\mathbb{C}} \setminus H$. Since $\widehat{\mathbb{C}} \setminus H \subset \widehat{\mathbb{C}} \setminus A_i$ and the configuration is unchanged, Y is simultaneously a computing triod in $\widehat{\mathbb{C}} \setminus A_i$. Reading the branch point of this one triod in both domains shows that the prime-end triple has the same cyclic orientation in $\widehat{\mathbb{C}} \setminus H$ as in $\widehat{\mathbb{C}} \setminus A_i$. Every ordered triple therefore retains its orientation, and the cyclic order of the whole family is unchanged. □

The realization theorem. Assembling the two lemmas yields the main result of this section.

Theorem 5.4 (Protected ribbon-routing realization; manuscript Theorem 10.4)**.** *Every protected ribbon-routing certificate R has an actual carrier H_R, given by* (10)*, with the following properties:*

1. *$H_R \Subset U$ is compact, connected, and full;*
2. *it contains every A_i, misses Z, and has singleton attachment trace $\{x_i\}$ at each atom;*
3. *every protected crosscut chain remains a prime-end chain in $\widehat{\mathbb{C}} \setminus H_R$;*

(4) *the survival map is injective, and it preserves the cyclic order of every protected family carried by a common source complement;*
(5) *every sufficiently thin regular neighborhood of the connector tree has the boundary word of the retained ribbon structure; in particular, connector-side readings made through such a neighborhood occur in the declared order; and*
(6) *no local connectivity of any A_i is required.*

Proof. *Clause* (1). As observed before Lemma 5.2, the set $T = G \cup \bigcup_i \alpha_i$ is a finite plane tree meeting each atom A_i in exactly the point x_i and otherwise missing every atom. Lemma 5.2 applies to T and $A_1, \dots, A_s$ and shows that $H_R = T \cup \bigcup_i A_i$ is a full continuum. Each constituent is compactly contained in U, and there are finitely many, so $H_R \Subset U$.

Clause (2). Containment of the atoms is part of (10). The certificate conditions of Definition 5.1 state that the arcs $\alpha_i((0,1])$ and the tree G miss Z, and Z misses every atom by hypothesis, so $H_R \cap Z = \varnothing$. The attachment trace at A_i is the intersection of A_i with the connector: $T \cap A_i = \{x_i\}$ because $\alpha_i \cap A_i = \{x_i\}$, the other port arcs and G miss A_i, and the atoms are pairwise disjoint.

Clause (3). Fix a protected chain $\Gamma_{i,a}$. We verify the hypothesis of Lemma 5.3 for $H = H_R$ and the atom A_i. The difference $H_R \setminus A_i$ is contained in

$$G \cup \bigcup_j \alpha_j((0,1]) \cup \bigcup_{j \neq i} A_j.$$

The tree G and every half-open port arc $\alpha_j((0,1])$ miss *all* relative first collar tails, in particular $\overline{D_{i,a,1}} \setminus A_i$, by Definition 5.1; and each atom A_j with $j \neq i$ misses the same relative collar tail by the truncation normalization. Hence

$$(H_R \setminus A_i) \cap (\overline{D_{i,a,1}} \setminus A_i) = \varnothing :$$

the connector misses the relative terminal collar tail, and therefore misses every deeper relative tail as well. Lemma 5.3 now shows that $\Gamma_{i,a}$ itself is the same prime-end chain in $\widehat{\mathbb{C}} \setminus H_R$.

Clause (4). The truncation normalization makes the relative terminal collar tails, hence the terminal selected domains, pairwise disjoint. For each atom A_i, the final part of Lemma 5.3 with $H = H_R$ gives distinctness and preservation of cyclic order for the subfamily of chains protecting A_i. The distinctness argument of the lemma uses only disjointness of the terminal selected domains and survival of both chains in $\widehat{\mathbb{C}} \setminus H_R$, so it applies verbatim to a pair of chains protecting different atoms as well. The survival map is therefore injective on the full family over all (i,a), and it preserves the cyclic order of every protected family with a common source domain.

Clause (5). Let N be a sufficiently thin regular neighborhood of the finite connector tree T in the plane; it can be chosen to miss Z because T and Z are disjoint compact sets. A regular neighborhood of a finite plane tree is a closed disk, and its boundary circle traverses every edge of T on two

sides: following ∂N, one passes along each edge once in each direction, and at a vertex of valence d the boundary visits the d half-edges in exactly their planar cyclic order. The resulting cyclic *boundary word*, in which every edge appears twice with opposite orientations, is precisely the boundary word of the ribbon structure retained in the certificate. This is the precise content of the connector-side order assertion: no additional prime end associated to an individual tree edge is asserted to exist.

Clause (6). The argument uses crosscut chains only: atoms enter through their prime-end chains and through the single point x_i, never through a boundary parametrization. No Euclidean boundary extension of an exterior Riemann map is assumed, and no local-connectivity hypothesis is imposed on an atom or on its boundary. □

Remark 5.5. Clause (6) is the point of the whole construction and deserves emphasis. Realization arguments for connectors frequently invoke a continuous boundary parametrization of a Riemann map, a blanket hypothesis equivalent by Carathéodory's theorem to local connectivity of the boundary. (Landing of one particular access is much weaker and is not conflated with local connectivity here.) The certificate formalism removes the blanket hypothesis: an access is a chain of crosscuts with declared sides, an object available for an arbitrary full continuum, and Lemma 5.3 shows it is preserved verbatim under any enlargement respecting the collar. This is what allows the framework to be applied to atoms that are not known—and are not claimed—to be locally connected.

Univalent naturality. The certificate data and the carrier construction are natural under univalent change of the Jordan localization.

Proposition 5.6 (Univalent naturality; manuscript Proposition 10.5)**.** *If $\phi\colon U \to V$ is an orientation-preserving conformal homeomorphism of Jordan disks and every certificate entry is transported by ϕ, then*

$$H_{\phi_* R} = \phi(H_R). \tag{11}$$

Choose any orientation-preserving sphere homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ extending ϕ. The protected survival map commutes, on the recorded protected chains, with the prime-end map induced by Φ, and the ribbon boundary order is preserved. These restricted correspondences are independent of the chosen extension because all recorded data lie in U.

Proof. Since U and V are Jordan disks, Carathéodory's extension theorem [14] extends ϕ to a homeomorphism $\overline{U} \to \overline{V}$ of the closed disks. By the Schoenflies theorem the restricted boundary homeomorphism $\partial U \to \partial V$ extends to a homeomorphism of the complementary closed disks $\widehat{\mathbb{C}} \setminus U \to \widehat{\mathbb{C}} \setminus V$, and gluing the two extensions along the boundary curves produces an orientation-preserving homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ with $\Phi|_U = \phi$.

A sphere homeomorphism preserves every notion entering Definition 5.1 and Theorem 5.4: it carries compact connected sets to compact connected

sets and complements to complements, hence preserves fullness; it carries open Jordan arcs with endpoints on a boundary to the same, hence crosscuts to crosscuts; it maps the two complementary sides of a crosscut to the two sides of the image crosscut, hence selected sides to selected sides; it carries null chains to null chains (spherical diameters are not literally preserved, but by uniform continuity of Φ the image crosscuts of a null chain again have diameters tending to zero); it preserves intersections, so the port conditions $\alpha_i \cap A_i = \{x_i\}$ and all disjointness requirements transport; and, being orientation preserving, it preserves the planar cyclic order of half-edges at the tree vertices and the cyclic order of prime ends. Consequently the transported data—the disk V, the atoms $\phi(A_i)$, the obstacle $\phi(Z)$, the chains $\Phi(\Gamma_{i,a})$, the ports $\phi \circ \alpha_i$, and the tree $\phi(G)$ with its transported ribbon structure—form a protected ribbon-routing certificate in V. The defining union (10) then transports term by term:

$$H_{\phi_* R} = \phi(G) \cup \bigcup_i \phi(\alpha_i) \cup \bigcup_i \phi(A_i) = \phi\Big(G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i\Big) = \phi(H_R),$$

which is (11). Because the chosen Φ maps $\widehat{\mathbb{C}} \setminus H_R$ onto $\widehat{\mathbb{C}} \setminus H_{\phi_* R}$ carrying null chains to null chains and preserving their division relation, it induces a bijection of prime ends under which the survival map of Lemma 5.3 for H_R corresponds to the survival map for $H_{\phi_* R}$, and the ribbon boundary words of Theorem 5.4(5) correspond letter by letter. Any two such extensions agree with ϕ on the protected crosscuts, selected sides, atoms, ports, and connector tree, so this correspondence on the recorded data does not depend on the extension. □

Thus an orientation-preserving conformal map of Jordan localizations transports every certificate and its realization. It does not identify the result with an independently constructed carrier.

With realization settled, the burden shifts to existence: when do the port arcs and the routing tree demanded by Definition 5.1 exist at all? Section 7 answers this question exactly, in terms of a single component of the frozen free domain Ω_{free} admitting simultaneous singleton ports to every required atom.

6. Attachment cohomology and the separation ledger

The realization theorem uses a finite tree meeting each atom at one point. There are two logically distinct requirements hidden in that choice: the connector itself has no holes, and every attachment trace is connected. The next theorem separates these requirements and gives an exact attachment ledger. An extra component of a trace creates a complementary component, whereas a complementary component of the trace itself fills one separation already carried by the connector. Thus attachments can create and destroy holes, but only through two explicitly measured trace defects.

For two nonseparating plane continua, the connected-intersection sufficiency is Janiszewski's theorem and the converse belongs to the classical Janiszewski–Mullikin theory [9, 11, 13, 17]. The form needed here keeps arbitrary initial connector cohomology, allows separating traces, records the homomorphism killed by each attachment, and iterates over a disjoint atom family. It also sits naturally in the classical unicoherence and multicoherence tradition [1, 5]: the term $c_i - 1$ is the classical multicoherence contribution of the decomposition $B_i = B_{i-1} \cup A_i$. The contribution here is its simultaneous Mayer–Vietoris bookkeeping and carrier application; we make no priority claim for the underlying contact-count invariant.

For a proper compact set $K \subset \widehat{\mathbb{C}}$ whose complement has finitely many components, define its *separation excess* by

$$\sigma(K) := \#\pi_0(\widehat{\mathbb{C}} \setminus K) - 1. \tag{12}$$

Thus $\sigma(K) = 0$ exactly when K is nonseparating. Since $\widehat{\mathbb{C}} \setminus K$ is open in a surface, it is locally path connected, so its connected components and path components agree. Alexander duality for compacta therefore identifies

$$\sigma(K) = \operatorname{rank} \check{H}^1(K; \mathbb{Z}). \tag{13}$$

We use the compact-pair Mayer–Vietoris sequence for Čech cohomology; both facts are recalled in the generality needed here in [15, Chapter 6].

Theorem 6.1 (Attachment exact sequence and separation ledger)**.** *Let $s \geq 1$, and let $C, A_1, \dots, A_s \subset \mathbb{C}$ be compact connected sets. Assume that the atoms A_i are pairwise disjoint full continua and that*

$$T_i := C \cap A_i$$

is nonempty. Put $H = C \cup \bigcup_i A_i$, and let $B_0 = C$ and $B_i = C \cup A_1 \cup \cdots \cup A_i$. For each i, the restriction homomorphism

$$r_i \colon \check{H}^1(B_{i-1}; \mathbb{Z}) \longrightarrow \check{H}^1(T_i; \mathbb{Z})$$

is surjective, and there is a natural short exact sequence

$$0 \longrightarrow \widetilde{\check{H}}^{\,0}(T_i; \mathbb{Z}) \xrightarrow{\delta_i} \check{H}^1(B_i; \mathbb{Z}) \longrightarrow \ker r_i \longrightarrow 0. \tag{14}$$

If T_i is nonseparating for every i, this specializes, noncanonically, to

$$\check{H}^1(H; \mathbb{Z}) \cong \check{H}^1(C; \mathbb{Z}) \oplus \bigoplus_{i=1}^{s} \widetilde{\check{H}}^{\,0}(T_i; \mathbb{Z}). \tag{15}$$

Suppose now that $\widehat{\mathbb{C}} \setminus C$ and every $\widehat{\mathbb{C}} \setminus T_i$ have finitely many components, and that T_i has finitely many connected components, say c_i. Then all groups in (14) *have finite rank, all complements $\widehat{\mathbb{C}} \setminus B_i$ have finitely many components, and*

$$\boxed{\sigma(H) = \sigma(C) + \sum_{i=1}^{s} \big(c_i - 1 - \sigma(T_i)\big).} \tag{16}$$

In particular, at the i-th attachment

$$\sigma(T_i) \le \sigma(B_{i-1}), \qquad \sigma(B_i) = \sigma(B_{i-1}) + c_i - 1 - \sigma(T_i). \tag{17}$$

Proof. Every B_i is connected. Indeed, $B_0 = C$ is connected, and at the i-th step the connected set A_i meets B_{i-1} in the nonempty set T_i. Pairwise disjointness of the atoms gives the exact intersection identity

$$B_{i-1} \cap A_i = (C \cap A_i) \cup \bigcup_{j<i} (A_j \cap A_i) = T_i.$$

Apply the reduced compact-pair Mayer–Vietoris sequence to the closed cover $B_i = B_{i-1} \cup A_i$. Its relevant segment is

$$\begin{aligned} \widetilde{\check{H}}^{\,0}(B_{i-1}) \oplus \widetilde{\check{H}}^{\,0}(A_i) &\longrightarrow \widetilde{\check{H}}^{\,0}(T_i) \xrightarrow{\delta_i} \check{H}^1(B_i) \\ &\longrightarrow \check{H}^1(B_{i-1}) \oplus \check{H}^1(A_i) \longrightarrow \check{H}^1(T_i). \end{aligned}$$

The first two reduced groups vanish because B_{i-1} and A_i are connected. Fullness of A_i and Alexander duality give $\check{H}^1(A_i;\mathbb{Z}) = 0$. The next term of the sequence is $\check{H}^2(B_i;\mathbb{Z})$, and it vanishes: B_i is a compact subset of the plane, hence a proper compact subset of S^2, and compacta Alexander duality identifies its reduced second cohomology with reduced homology in degree -1 of the nonempty complement. Exactness therefore makes r_i surjective and leaves exactly (14).

Alexander duality identifies $\check{H}^1(B_{i-1};\mathbb{Z})$ with the reduced zeroth homology of the open surface $\widehat{\mathbb{C}} \setminus B_{i-1}$, hence with a free abelian group. Its subgroup $\ker r_i$ is free and therefore projective, so (14) splits, although not canonically. When every T_i is nonseparating, the target of r_i vanishes and iteration gives (15).

Under the finiteness hypotheses, the components of T_i are clopen and

$$\operatorname{rank} \widetilde{\check{H}}^{\,0}(T_i;\mathbb{Z}) = c_i - 1, \qquad \operatorname{rank} \check{H}^1(T_i;\mathbb{Z}) = \sigma(T_i).$$

Surjectivity of r_i gives the first inequality in (17); taking ranks in (14) gives its equality. Starting from the finite-rank group $\check{H}^1(C;\mathbb{Z})$, induction makes every $\check{H}^1(B_i;\mathbb{Z})$ finite rank. Alexander duality then makes every complement $\widehat{\mathbb{C}} \setminus B_i$ have finitely many components. Summing the step equalities gives (16). □

The ledger has a useful geometric reading. For example, let C be the unit circle together with the radial segment $[1,2]$, and attach the full closed unit disk A. The trace $C \cap A$ is the unit circle. It is connected but has separation excess one, so the attachment fills the one bounded complementary component of C:

$$\sigma(C \cup A) = 1 + (1 - 1) - 1 = 0.$$

In contrast, a nonseparating trace cannot cancel connector homology; its only possible contribution is the creation term $c_i - 1$.

The trace-separation term vanishes automatically for the carrier geometry that matters most here.

Lemma 6.2 (Intersections of full compacta are nonseparating)**.** *If $K, L \subset \mathbb{C}$ are compact and full, then $K \cap L$ is nonseparating (whether or not it is connected). In particular $\check{H}^1(K \cap L; \mathbb{Z}) = 0$.*

Proof. If $K \cap L = \varnothing$, its complement is $\mathbb{C}$ and $\check{H}^1(\varnothing; \mathbb{Z}) = 0$, so the conclusion is immediate. Assume $K \cap L \neq \varnothing$. The two connected sets $\mathbb{C} \setminus K$ and $\mathbb{C} \setminus L$ contain a common neighborhood of infinity. Their union is therefore connected, and

$$\mathbb{C} \setminus (K \cap L) = (\mathbb{C} \setminus K) \cup (\mathbb{C} \setminus L)$$

is connected. Alexander duality gives the cohomological conclusion. □

Corollary 6.3 (Exact criterion for a full continuum connector)**.** *In Theorem 6.1, suppose that C is also a full continuum. Then every trace is nonseparating, and H is full if and only if every attachment trace $C \cap A_i$ is connected. If, in addition, T_i has finitely many connected components c_i for every i, then*

$$\#\pi_0(\widehat{\mathbb{C}} \setminus H) = 1 + \sum_{i=1}^{s} (c_i - 1). \tag{18}$$

Proof. Lemma 6.2, applied to C and A_i, verifies that $\check{H}^1(T_i; \mathbb{Z}) = 0$. The splitting (15), together with $\check{H}^1(C; \mathbb{Z}) = 0$, gives

$$\check{H}^1(H; \mathbb{Z}) \cong \bigoplus_i \widetilde{\check{H}}^0(T_i; \mathbb{Z}).$$

For a compact Hausdorff space, reduced Čech cohomology in degree zero vanishes exactly when the space is connected. Alexander duality now gives the stated if-and-only-if criterion without a finiteness assumption. When all c_i are finite, every T_i has $\sigma(T_i) = 0$ and $\sigma(C) = 0$, so (16) becomes (18). □

Corollary 6.3 both strengthens and sharpens the one-point tree formulation of Lemma 5.2, and its if-and-only-if clause recovers the Janiszewski–Mullikin criterion in this finite-trace setting. The connector need not be a tree, locally connected, or one dimensional: any full continuum is allowed. Singleton traces can be replaced by arbitrary connected traces. Conversely, disconnected traces are not merely inconvenient. When the connector is full they force holes, and the number forced is exact.

Corollary 6.4 (Protected realization by a continuum connector)**.** *Retain the Jordan localization, atoms, obstacle set, and protected crosscut chains of Section 5, with pairwise disjoint relative terminal collar tails. Let $C \Subset U$ be a full continuum such that:*

(1) $C \cap Z = \varnothing$*;*
(2) $C \cap A_i$ *is nonempty and connected for every i; and*
(3)

$$(C \setminus A_i) \cap \left(\overline{D_{i,a,1}} \setminus A_i\right) = \varnothing \quad \text{for every protected chain } (i, a).$$

Then

$$H_C := C \cup \bigcup_i A_i$$

is a compact connected full carrier avoiding Z. Every protected chain survives in $\widehat{\mathbb{C}} \setminus H_C$; the survival assignment is injective on families with disjoint relative terminal collar tails and preserves the cyclic order carried by each common source complement. No local connectivity of C or of any atom is required.

Proof. Corollary 6.3 makes H_C full, and the nonempty traces make it connected. The collar-clearance hypothesis for each chain is exactly the hypothesis of Lemma 5.3, because the other atoms miss its relative terminal collar tail. That lemma supplies survival, injectivity, and cyclic-order preservation. Compact containment and obstacle avoidance follow from the stated hypotheses and the finite union. □

6.1. **Sharpness and defect birth at a limit.** The separation ledger exposes a compactness obstruction that is invisible at every finite stage. Even a sequence of tree carriers with singleton traces can converge to a carrier with arbitrarily many bounded complementary components.

Proposition 6.5 (Arbitrary separation defect born in a Hausdorff limit)**.** *For every integer $m \geq 1$ there are a full atom A, full arc connectors C_n, and a full arc C such that*

$$C_n \longrightarrow C \quad \text{and} \quad A \cup C_n \longrightarrow A \cup C$$

in the Hausdorff metric,

$$A \cap C_n \text{ is a singleton for every } n, \qquad \#(A \cap C) = m + 1,$$

every $A \cup C_n$ is full, but $\widehat{\mathbb{C}} \setminus (A \cup C)$ has exactly $m+1$ components. Thus the limiting carrier has exactly m bounded complementary components.

Proof. Take

$$A = [0, m] \times \{0\}.$$

For $n \geq 1$, let C_n be the polygonal arc whose ordered vertices are

$$(0,0),\ \left(\tfrac{1}{2}, 1\right),\ (1, n^{-1}),\ \left(\tfrac{3}{2}, 1\right),\ (2, n^{-1}),\ \ldots,\ \left(m - \tfrac{1}{2}, 1\right),\ (m, n^{-1}).$$

The first coordinate increases strictly along the vertex list, so C_n is a Jordan arc. It lies in the closed upper half-plane and meets the horizontal atom A only at $(0,0)$. Both A and C_n are full, and their intersection is connected; hence $A \cup C_n$ is full, either by Janiszewski's theorem or by Corollary 6.3.

Let C be the polygonal arc obtained by replacing every vertex (j, n^{-1}) by $(j, 0)$. The vertex convergence, with the same finite linear parametrization, gives $C_n \to C$ in the Hausdorff metric and hence $A \cup C_n \to A \cup C$. The limit arc is still x-monotone and

$$A \cap C = \{(0,0), (1,0), \ldots, (m,0)\},$$

which has $m+1$ components. Apply Corollary 6.3 with connector C and atom A:

$$\#\pi_0\big(\widehat{\mathbb{C}} \setminus (A \cup C)\big) = 1 + \big((m+1) - 1\big) = m+1.$$

The component containing infinity is the unique unbounded component, so the other m components are bounded. □

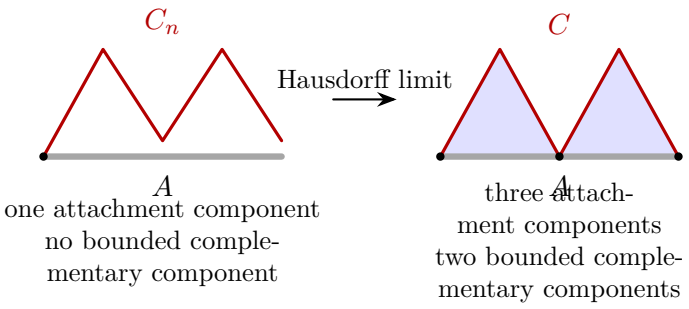


FIGURE 5. Trace multiplication in Proposition 6.5, shown for $m = 2$. Every finite connector C_n meets the atom once, so $A \cup C_n$ is full. In the limit, two near-contacts close and the trace $A \cap C$ has three components. Formula (18) detects the two shaded bounded complementary components exactly.

6.2. **A safe limiting regime.** The preceding defect is specific to uncontrolled Hausdorff convergence. There is a useful inverse-limit regime in which neither connector holes nor trace components can be born.

Theorem 6.6 (Nested-limit stability)**.** *Let $s \geq 1$, let $A_1, \dots, A_s \subset \mathbb{C}$ be pairwise disjoint full continua, and let*

$$C_1 \supset C_2 \supset \cdots$$

be a decreasing sequence of nonempty full continua. Suppose that $C_n \cap A_i$ is nonempty and connected for every n and i. Then

$$C_\infty := \bigcap_{n \geq 1} C_n$$

is a full continuum, every trace $C_\infty \cap A_i$ is nonempty and connected, and

$$H_\infty := C_\infty \cup \bigcup_{i=1}^{s} A_i$$

is a full continuum.

If, in addition, the data lie in the setup of Corollary 6.4, every $C_n \Subset U$ avoids the fixed obstacle set, and every C_n satisfies that corollary's relative-collar clearance conditions, then H_∞ inherits obstacle avoidance and every protected prime-end chain survives with the same injectivity and cyclic-order conclusions.

Proof. The nested-intersection theorem for continua makes C_∞ nonempty, compact, and connected. Moreover

$$\widehat{\mathbb{C}} \setminus C_\infty = \bigcup_{n\geq 1} (\widehat{\mathbb{C}} \setminus C_n).$$

The sets on the right are connected and increasing, and all contain ∞; their union is connected. Hence C_∞ is full.

For fixed i, the continua $C_n \cap A_i$ are nonempty, compact, connected, and decreasing. Their intersection is therefore nonempty and connected, and distributivity of intersection gives

$$\bigcap_n (C_n \cap A_i) = C_\infty \cap A_i.$$

Corollary 6.3 now makes H_∞ full. Avoidance and relative-collar clearance pass to the subset C_∞; the final assertion follows from Corollary 6.4. □

The theorem suggests a concrete route for an infinite construction: produce a *decreasing* system of full connector envelopes with connected, nonempty traces, rather than take an arbitrary subsequential Hausdorff limit of finite trees. It is a sufficient architecture, not a claim that the dynamical program already supplies such envelopes.

Warning 6.7 (The compactness gate)**.** Fullness and singleton attachment traces are not closed under Hausdorff limits. Proposition 6.5 isolates the trace-birth failure, while

$$C_n = \{e^{it} : 1/n \leq t \leq 2\pi\}$$

are closed arcs converging in Hausdorff distance to the separating Jordan curve S^1, isolating the independent cycle-birth failure in the connector itself. Therefore a finite protected-carrier theorem cannot be passed to an infinite or virtual horizon by compactness alone. At minimum, one must prove that the limiting connector remains nonseparating and that each limiting attachment trace remains nonempty and connected. Under exactly those two conclusions, Corollary 6.3 restores fullness of the limiting carrier.

The warning is deliberately a no-go statement rather than a promoted dynamical claim. It sharpens the finite/infinite boundary of this paper: the finite certificate solves routing, but a future limiting construction must control both connector cohomology and trace multiplication. Neither control follows from Hausdorff precompactness. Theorem 6.6 supplies one sufficient replacement: decreasing full envelopes with nonempty connected traces.

7. The exact free-component criterion

Theorem 5.4 converts a protected ribbon-routing certificate into an actual carrier, but it leaves open the question of when such a certificate exists: the embedded finite tree of Definition 5.1 looks like an arbitrary choice, and one might fear that its existence depends on delicate features of the atoms

or of the forbidden data. The present section removes the arbitrary tree choice. We prove that the existence of a protected ribbon-routing certificate is governed by a single, exact, purely planar condition: one component of a frozen free domain must reach every required atom by a singleton port. The section carries the one intentional mathematical divergence of this paper from the source manuscript [12]: the slit lemma below is stated and proved in a repaired form, and the proof of the criterion is rewritten to match. The divergence is recorded precisely in Remark 7.2.

7.1. **The frozen free domain.** Freeze the complete finite future inventory before choosing a connector. Concretely, fix $s \geq 1$, let U be the Jordan localization, let $A_1, \dots, A_s \Subset U$ be the pairwise disjoint atoms, and let $Z \Subset U \setminus \bigcup_i A_i$ be the same compact set of separated obstacles, critical-value exclusions, and localization-forbidden data used in Definition 5.1. For each atom define its compact *atom–tail block*

$$\mathcal{B}_i := A_i \cup \bigcup_a (\overline{D_{i,a,1}} \setminus A_i) = A_i \cup \bigcup_a \overline{D_{i,a,1}},$$

and put

$$Q := \bigcup_{i=1}^{s} \mathcal{B}_i \cup Z.$$

Thus every relative tail is processed together with its atom, even when different prime-end impressions overlap inside A_i. Each $\overline{D_{i,a,1}}$ is connected and meets A_i in its nonempty impression, so $\mathcal{B}_i$ is a compact connected set (and equals A_i when there are no protected labels). The truncation normalization and compact containment of Z make the finite union $Q \Subset U$ compact. Put

$$\Omega_{\mathrm{free}} \;=\; U \setminus Q \tag{19}$$

(equation (9.4) of the manuscript [12]). The set Ω_{free} is open, and its connected components are plane domains, the *free components* of the frozen data.

A component Ω_0 of Ω_{free} *has simultaneous singleton ports* to the required atoms if there are pairwise disjoint Jordan arcs $\alpha_1, \dots, \alpha_s \colon [0,1] \to U$ such that, for every i, the arc α_i meets A_i exactly in the single point $x_i = \alpha_i(0)$ and satisfies $\alpha_i\big((0,1]\big) \subset \Omega_0$. The landing point x_i is called a *free point* of A_i because it is reached from the free domain; no further property of x_i is assumed at this stage. (Compare the unprotected exposure loci of Lemma 9.2, where an additional protected-impression exclusion is imposed; the interaction between the two notions is illustrated in Example 7.4 below.)

The adjective *frozen* is load-bearing, and we emphasize it once. The set Q lists the complete finite future inventory: every atom that will ever have to be joined, the relative terminal collar tail of every protected chain that any future step of a proof will read, every obstacle that must remain separated, the full critical-value exclusion, and the localization-forbidden set. This list is fixed *before* any connector is chosen. Nothing may be added to Q

afterwards, and nothing may be removed from it; all later monotone operations act on proposed connectors, never on the inventory. This discipline is exactly what gives the criterion of Theorem 7.3 its if-and-only-if form, and what makes its failure clause persistent: a separation already visible in the frozen data can never be talked away by enlarging a connector. The horizon-first closure results of the manuscript [12] address how such an inventory stabilizes; here the inventory is simply given, finite, and immutable.

7.2. **The repaired slit lemma.** The sufficiency direction of the criterion rests on the fact that finitely many pairwise disjoint one-ended slits do not disconnect a plane domain. This is Lemma 10.7 of the manuscript. Its printed statement contains a formal inconsistency (a map into an open domain cannot take a boundary value) together with an endpoint mismatch with its intended application; both defects are repaired in the statement below, and Remark 7.2 records the divergence. The mathematical core of the manuscript's argument — the transverse-rectangle detour around the free endpoint — is retained in full.

Lemma 7.1 (One-ended slits do not disconnect a plane domain; manuscript Lemma 10.7, repaired)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\delta_1, \dots, \delta_s \colon [0,1] \to \overline{\Omega}$ be Jordan arcs with pairwise disjoint images such that $\delta_i(0) \in \partial\Omega$ and $\delta_i\big((0,1]\big) \subset \Omega$ for every i. Then*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i\big((0,1)\big)$$

is path connected; in particular every free endpoint $\delta_i(1)$ lies in this set and can be joined inside it to every other point of it.

Proof. Call δ_i the *i-th slit*, write $\sigma_i := \delta_i\big((0,1]\big)$ for its *half-open* part and $\sigma_i^\circ := \delta_i\big((0,1)\big)$ for its *open* part, and call $\delta_i(1)$ its *tip*. The set to be proved path connected is $\Omega \setminus \bigcup_i \sigma_i^\circ$, which contains all s tips. If $s = 0$ there is nothing to prove, so assume $s \geq 1$; then $\partial\Omega \neq \varnothing$.

Step 0: two preliminary observations. First, each half-open part σ_i is relatively closed in Ω. Indeed, the closure of σ_i in $\mathbb{C}$ is the compact arc $\delta_i([0,1])$, so the only point of $\overline{\sigma_i} \setminus \sigma_i$ is $\delta_i(0) \in \partial\Omega$, which does not lie in Ω. Consequently $\Omega \setminus \sigma_i$ is open. By contrast, the open part σ_i° is *not* relatively closed in Ω: its closure inside Ω also contains the tip $\delta_i(1) \in \Omega$. This distinction is precisely the point at which the manuscript's induction required repair; the proof below removes the half-open parts first and reattaches the tips afterwards.

Second, every Jordan arc in the plane is tame: it can be extended to a Jordan curve, and the Schoenflies theorem then supplies a homeomorphism of $\mathbb{C}$ carrying the curve to a round circle and the arc to a circular subarc, hence, after one more homeomorphism, onto the straight segment $[0,1]\times\{0\}$; see, e.g., [13, 17]. Since path connectivity is a topological invariant and a

homeomorphism of $\mathbb{C}$ carries domains to domains and boundaries to boundaries, we may straighten any *single* slit whenever convenient. (Straightening one slit will in general bend the others; each step below straightens only the slit it is processing.)

Step 1: removing a single half-open slit leaves a domain. Let Ω be a domain and let δ be a single arc as in the statement, with half-open part $\sigma = \delta\big((0,1]\big)$. We claim that $\Omega' := \Omega \setminus \sigma$ is again a domain. Openness was proved in Step 0, so only path connectivity is at issue. By tameness we may assume, after a homeomorphism of $\mathbb{C}$, that

$$\delta(t) = (t, 0) \quad (0 \le t \le 1),$$

so that Ω is a domain containing $S := (0,1] \times \{0\}$, with $(0,0) \in \partial\Omega$, and we must join two arbitrary points $x, y \in \Omega \setminus S$ by a path in $\Omega \setminus S$.

(a) A polygonal start missing an initial tail. Since a plane domain is polygonally connected [13], choose a polygonal path $P \subset \Omega$ from x to y. The path P is compact and does not contain the boundary point $(0,0)$, so

$$r := \operatorname{dist}\big((0,0), P\big) > 0.$$

Fix $\varepsilon := \frac{1}{2}\min(r, 1)$. Every point of the initial tail $[0, \varepsilon] \times \{0\}$ lies within distance $\varepsilon < r$ of the origin, so P misses $[0, \varepsilon] \times \{0\}$. This is the only part of the closed slit that a compact subset of Ω cannot in general avoid by compactness alone: the closed slit image leaves every compact subset of Ω only along points accumulating at $\delta(0) = (0,0) \in \partial\Omega$, and near that end the continuity of δ at 0 confines the slit to a small disk around the origin. The remaining part $\delta([\varepsilon, 1]) = [\varepsilon, 1] \times \{0\}$ is a compact subset of Ω and will be handled by explicit detours.

(b) A thin rectangle around the compact subarc. The segment $[\varepsilon, 1] \times \{0\}$ lies in Ω, and $(1,0) \in \Omega$ is an interior point of Ω; hence there is $\eta \in (0, \frac{1}{2})$ with $[1, 1+\eta] \times \{0\} \subset \Omega$. The segment $L := [\varepsilon/2,\, 1+\eta] \times \{0\}$ is then a compact subset of Ω, so $\operatorname{dist}(L, \mathbb{C} \setminus \Omega) > 0$, and we may choose $h \in \big(0, \min(\varepsilon/2, \operatorname{dist}(L, \mathbb{C} \setminus \Omega)/2)\big)$ and set

$$R := [\varepsilon/2,\, 1+\eta] \times [-h, h] \subset \Omega.$$

The rectangle R meets the closed slit $[0,1] \times \{0\}$ exactly in the core $c := [\varepsilon/2,\, 1] \times \{0\}$: the slit enters R through the midpoint of its left short side and terminates at the tip $(1,0)$, an interior point of R. In particular $R \cap S = [\varepsilon/2, 1] \times \{0\} = c$.

We record the key connectivity property of the slit rectangle: *$R \setminus c$ is path connected.* Indeed, let $V := \{1 + \eta/2\} \times [-h, h]$, a vertical segment in R missing c. A point of $R \setminus c$ with ordinate $y \neq 0$ travels horizontally to the point of V with the same ordinate; the horizontal path misses the axis $\mathbb{R} \times \{0\}$ entirely, hence misses c. A point of $R \setminus c$ with ordinate 0 has abscissa in $(1, 1+\eta]$ and travels along the axis to $(1+\eta/2, 0)$; this path keeps abscissa > 1, hence misses c. Since V is connected, the claim follows. Geometrically:

the two long sides of R are joined to each other inside $R \setminus c$ *around the tip*, by crossing the axis at abscissae beyond 1.

(c) Transverse perturbation and detour repair. Perturb the interior vertices of P, keeping the endpoints x, y fixed, so that no interior vertex lies on the axis $\mathbb{R} \times \{0\}$ and no edge of P is collinear with the axis; if an endpoint happens to lie on the axis (necessarily off S), perturb its adjacent vertex so that the incident edge is not horizontal. Since P is compact with $\operatorname{dist}(P, \mathbb{C} \setminus \Omega) > 0$ and $\operatorname{dist}\big(P, [0,\varepsilon] \times \{0\}\big) \geq r - \varepsilon > 0$, a sufficiently small such perturbation keeps P inside Ω and away from the initial tail. After the perturbation, every edge of P meets the axis in at most one point, and every meeting point interior to an edge is a transverse crossing. Hence P meets S in finitely many points, all of the form $P(t_j)$ with parameters $0 < t_1 < \cdots < t_m < 1$ interior to edges, and all lying in $(\varepsilon, 1] \times \{0\} \subset \operatorname{int} R$. (If $m = 0$, the path P already joins x to y in $\Omega \setminus S$ and Step 1 is complete.)

By transversality, for each j there is $\rho_j > 0$ such that the parameter interval $[t_j - \rho_j,\, t_j + \rho_j]$ has image in $\operatorname{int} R$, meets the axis only at t_j, and the two punctured halves of the image lie strictly on opposite sides of the axis; shrink the ρ_j so that these intervals are pairwise disjoint and avoid 0 and 1. Put $p_j^- := P(t_j - \rho_j)$ and $p_j^+ := P(t_j + \rho_j)$; both lie in $R \setminus c$. Now modify P as follows: on each interval $[t_j - \rho_j,\, t_j + \rho_j]$, replace the sub-path of P by a path from p_j^- to p_j^+ inside $R \setminus c$, which exists by (b); keep P unchanged elsewhere. Each detour lies in $R \subset \Omega$ and misses S, because $S \cap R = c$ and the detour misses c, while $S \setminus R$ is disjoint from R. The unchanged pieces of P carry parameters outside the intervals $(t_j - \rho_j,\, t_j + \rho_j)$ and therefore avoid S, since the crossings t_j were the only parameters at which P met S. The concatenation is a path from x to y in $\Omega \setminus S$.

Observe that each detour crosses the axis only at abscissae strictly beyond 1: it travels *around* the tip, through points arbitrarily close to $\delta(1)$ but never through any point of the closed slit; the tip itself, which here has been removed together with the rest of σ, is not needed as a way-station. This completes Step 1: $\Omega' = \Omega \setminus \sigma$ is a domain.

Step 2: removing all half-open slits leaves a domain. Set $\Omega_{(0)} := \Omega$ and, inductively, $\Omega_{(j)} := \Omega_{(j-1)} \setminus \sigma_j$ for $1 \leq j \leq s$. We claim that each $\Omega_{(j)}$ is a domain and that δ_j is a one-ended slit of $\Omega_{(j-1)}$ in the sense of the statement. Indeed, suppose $\Omega_{(j-1)}$ is a domain. Because the slit images are pairwise disjoint, σ_j is disjoint from $\sigma_1, \ldots, \sigma_{j-1}$, so $\sigma_j \subset \Omega_{(j-1)}$. The point $\delta_j(0)$ lies outside $\Omega \supset \Omega_{(j-1)}$, and it is the limit of the points $\delta_j(t) \in \Omega_{(j-1)}$ as $t \downarrow 0$; hence $\delta_j(0) \in \partial\Omega_{(j-1)}$ and $\delta_j \colon [0,1] \to \overline{\Omega_{(j-1)}}$. Step 1, applied to the domain $\Omega_{(j-1)}$ and the single slit δ_j, now shows that $\Omega_{(j)}$ is a domain. After s steps we conclude: the set

$$\Omega_{(s)} \;=\; \Omega \setminus \bigcup_{i=1}^{s} \sigma_i \;=\; \Omega \setminus \bigcup_{i=1}^{s} \delta_i\big((0,1]\big)$$

is open and path connected.

Step 3: reattaching the tips. The set whose path connectivity is asserted differs from $\Omega_{(s)}$ exactly by the s tips:

$$\Omega \setminus \bigcup_{i=1}^{s} \sigma_i^\circ \;=\; \Omega_{(s)} \cup \{\delta_1(1), \dots, \delta_s(1)\},$$

and the tips are pairwise distinct points not in $\Omega_{(s)}$. Since $\Omega_{(s)}$ is path connected, it suffices to join each tip to some point of $\Omega_{(s)}$ by a path which, apart from the tip itself, lies in $\Omega_{(s)}$.

Fix i and straighten the i-th slit: let Φ be a homeomorphism of $\mathbb{C}$ with $\Phi\big(\delta_i([0,1])\big) = [0,1] \times \{0\}$ and $\Phi(\delta_i(t)) = (t, 0)$. The union $K := \bigcup_{k \neq i} \delta_k([0,1])$ of the other closed slit images is compact and disjoint from $\delta_i([0,1])$, so $\Phi(K)$ is compact and disjoint from $[0,1] \times \{0\}$. The model tip $(1,0)$ lies in the open set $\Phi(\Omega)$ and outside the closed set $\Phi(K)$, so

$$\mu := \operatorname{dist}\Big((1,0),\, \Phi(K) \cup \big(\mathbb{C} \setminus \Phi(\Omega)\big)\Big) > 0.$$

Choose $a \in \big(0, \min(\mu/2, \tfrac{1}{2})\big)$ and consider the square $R_i := [1-a,\, 1+a] \times [-a, a]$. Every point of R_i lies within distance $a\sqrt{2} < \mu$ of the model tip, so $R_i \subset \Phi(\Omega)$ and $R_i \cap \Phi(K) = \varnothing$; moreover R_i meets the model slit $[0,1] \times \{0\}$ exactly in $[1-a, 1] \times \{0\}$. The *escape segment*

$$E_i := [1,\, 1 + a/2] \times \{0\}$$

runs from the model tip into R_i; every point of E_i other than the tip has abscissa > 1, hence lies on none of the straightened slits, and lies in R_i, hence off $\Phi(K)$. Pulling back, $\beta_i := \Phi^{-1}(E_i)$ is a Jordan arc from $\delta_i(1)$ to a point $z_i \in \Omega_{(s)}$ with $\beta_i \setminus \{\delta_i(1)\} \subset \Omega_{(s)}$. Thus the tip $\delta_i(1)$ lies in the same path component as z_i, hence in the path component of $\Omega_{(s)}$.

The same square makes the two-sided accessibility of the tip transparent: inside $R_i \setminus \big([1-a, 1] \times \{0\}\big)$ the two long sides of R_i are joined around the tip by the vertical segment at abscissa $1 + a/2$, so the tip is accessible from either side of the slit, and both accesses land in one and the same path component of the complement. Finally, note that the escape arc β_i can be made arbitrarily small: shrinking a shrinks E_i to the model tip, and continuity of Φ^{-1} then confines β_i to any prescribed neighborhood of $\delta_i(1)$.

Step 4: conclusion. By Steps 2 and 3 the set $\Omega \setminus \bigcup_i \sigma_i^\circ = \Omega_{(s)} \cup \{\delta_1(1), \dots, \delta_s(1)\}$ is path connected, and it contains every free endpoint $\delta_i(1)$; each of these can therefore be joined inside it to every other point of it. □

Two by-products of the proof will be used in the proof of Theorem 7.3 and are worth recording explicitly. First, by Step 2, removing the *half-open* slits $\delta_i((0,1])$ leaves an open, path connected set — indeed a plane domain, hence a polygonally connected one. Second, by Step 3, every tip $\delta_i(1)$ admits arbitrarily small escape arcs: for every neighborhood N of $\delta_i(1)$ there is a Jordan arc $\beta \subset N$ from $\delta_i(1)$ into the slit complement with $\beta \setminus \{\delta_i(1)\} \subset \Omega \setminus \bigcup_k \delta_k\big((0,1]\big)$.

Remark 7.2 (The repair). We record the divergence from the printed manuscript statement [12]. Manuscript Lemma 10.7 states the hypothesis in the form $\delta_i\colon [0,1] \to \Omega$ with $\delta_i(0) \in \partial\Omega$, which is formally inconsistent: a map into the open domain Ω cannot take a boundary value. It then removes the half-open slits $\delta_i\big((0,1]\big)$, *including* the free endpoints $\delta_i(1)$, while the manuscript's proof of Theorem 10.8 (reproduced as Theorem 7.3 below) joins precisely those outer endpoints by paths inside the complement — an endpoint mismatch: the points to be joined have just been deleted. The statement of Lemma 7.1 repairs both defects at once: the codomain is $\overline{\Omega}$, so the initial boundary value is legitimate, and only the open arcs $\delta_i\big((0,1)\big)$ are removed, so the free endpoints remain available for routing.

A third, smaller point is repaired inside the proof. The manuscript's induction asserts that the removed slit is relatively closed in Ω, so that the complement is again a domain and the one-slit argument iterates. For the half-open slit $\delta_i\big((0,1]\big)$ this is true (Step 0), but for the open slit $\delta_i\big((0,1)\big)$ it is false: the closure inside Ω adds the tip, so the complement of the open slit is not open at the tip and the inductive step cannot be applied verbatim to the repaired statement. The proof above therefore removes the half-open slits one at a time (Steps 1 and 2), where relative closedness does hold, and reattaches the s retained tips afterwards by the tip-accessibility argument (Step 3).

The repair changes no application. Every invocation of the lemma in the source manuscript — and every invocation in this paper — uses exactly the repaired form: the slits to be removed are open port interiors, and the retained outer endpoints are then joined inside the complement. The transverse-rectangle detour, which is the mathematical core of the manuscript's one-slit argument, is unchanged.

7.3. **The criterion.**

Theorem 7.3 (Protected Free-Component Criterion; manuscript Theorem 10.8)**.** *A protected ribbon-routing certificate exists if and only if one component of Ω_{free} has simultaneous singleton ports to every required atom. Consequently, if Q is fixed and no component has such ports, no enlargement of a proposed connector that continues to respect the same frozen inventory can become a protected ribbon-routing certificate.*

Proof. Necessity. Let a protected ribbon-routing certificate be given, with free-port arcs $\alpha_1, \dots, \alpha_s$, landing points $x_i = \alpha_i(0) \in A_i$, and embedded finite tree G joining the outer port endpoints (Definition 5.1). Delete the atoms from the connector and consider

$$X \;:=\; G \cup \bigcup_{i=1}^{s} \alpha_i\big((0,1]\big).$$

The tree G is connected, and each half-open port arc $\alpha_i\big((0,1]\big)$ is a connected set containing the outer endpoint $\alpha_i(1) \in G$; hence X is connected.

Moreover $X \subset \Omega_{\mathrm{free}}$: by Definition 5.1, the sets $\alpha_i\big((0,1]\big)$ and G lie in U and avoid every atom, the obstacle and critical-value set, and every relative terminal protected collar tail. The localization-forbidden data are part of Z by the convention of Section 5. The avoided sets are therefore exactly the constituents of Q. A connected subset of the open set Ω_{free} lies in a single component Ω_0. The certificate's own port arcs are then simultaneous singleton ports of Ω_0: they are pairwise disjoint Jordan arcs, α_i meets A_i exactly in $\{x_i\}$, and $\alpha_i\big((0,1]\big) \subset X \subset \Omega_0$.

Sufficiency. Conversely, let Ω_0 be a component of Ω_{free} with simultaneous singleton ports $\alpha_1, \dots, \alpha_s$. Fix $\tau \in (0,1]$ and shorten each port away from the atoms, replacing α_i by its initial subarc $\alpha_i|_{[0,\tau]}$, reparametrized over $[0,1]$ and still denoted α_i; the shortened arcs remain pairwise disjoint simultaneous singleton ports, and the shortening emphasizes that only the germ of each port at its atom matters. Write $y_i := \alpha_i(1)$ for the retained outer endpoints.

The shortened ports are one-ended slits of the domain Ω_0 in the sense of Lemma 7.1. Indeed, their images are pairwise disjoint; the interiors satisfy $\alpha_i\big((0,1]\big) \subset \Omega_0$; and the landing point $x_i = \alpha_i(0)$ lies in $A_i \subset Q$, hence outside Ω_0, while being the limit of the points $\alpha_i(t) \in \Omega_0$, so that $x_i \in \partial\Omega_0$ and $\alpha_i \colon [0,1] \to \overline{\Omega_0}$. The retained port interiors are therefore finitely many pairwise disjoint one-ended slits of Ω_0, and Lemma 7.1 applies to remove their *open* parts: the set

$$W \;:=\; \Omega_0 \setminus \bigcup_{i=1}^{s} \alpha_i\big((0,1)\big)$$

is path connected, and it contains the outer endpoints $y_1, \dots, y_s$ — which is exactly what the repaired statement guarantees.

We now upgrade the joining paths supplied by path connectivity of W to an embedded polygonal tree, using the two by-products recorded after the proof of Lemma 7.1. First, by Step 2 of that proof,

$$\Omega' \;:=\; \Omega_0 \setminus \bigcup_{i=1}^{s} \alpha_i\big((0,1]\big)$$

is a plane domain. Second, by Step 3, each tip y_i admits arbitrarily small escape arcs into Ω'. If $s = 1$, choose such an escape arc β_1 from y_1 to a point $z_1 \in \Omega'$, replace the port by $\alpha_1 * \beta_1$, and take the degenerate one-vertex tree $G = \{z_1\}$. The incidence and avoidance clauses are then immediate, so this is a certificate. Assume henceforth that $s \geq 2$. Let $\rho := \frac{1}{3}\min_{i \neq j} |y_i - y_j| > 0$, and choose for each i an escape arc $\beta_i \subset B(y_i, \rho)$ with $\beta_i(0) = y_i$ and $\beta_i\big((0,1]\big) \subset \Omega'$; the balls $B(y_i, \rho)$ are pairwise disjoint, so the arcs β_i are pairwise disjoint. Put $z_i := \beta_i(1) \in \Omega'$.

Since the domain Ω' is polygonally connected, choose polygonal paths $P_2, \dots, P_s \subset \Omega'$ joining z_1 to $z_2, \dots, z_s$. Perturb them, keeping their endpoints fixed, into general position inside Ω': after an arbitrarily small perturbation, any two edges taken from distinct paths, or nonadjacent edges of one path, meet in at most one point and transversely, and no vertex of one path lies on another path except at shared endpoints. The union $\Gamma := P_2 \cup \cdots \cup P_s$ is then a connected finite plane graph in Ω' containing $z_1, \dots, z_s$: its vertices are the path endpoints together with the finitely many transverse crossings, at which we subdivide. Let $G_0 \subset \Gamma$ be an inclusion-minimal connected subgraph containing $\{z_1, \dots, z_s\}$. Then G_0 is a finite tree: were it to contain a cycle, deleting one edge of the cycle would leave a connected subgraph still containing every z_i, contradicting minimality. Being a subgraph of an embedded graph, G_0 is embedded, and $G_0 \subset \Omega'$.

The escape arcs may meet G_0 before their endpoints, so we trim them by a first-hit argument. For each i the set $\{t \in [0,1] : \beta_i(t) \in G_0\}$ is closed, and it is nonempty because $\beta_i(1) = z_i \in G_0$; let t_i^* be its minimum. Since $\beta_i(0) = y_i \notin \Omega' \supset G_0$ we have $t_i^* > 0$. Set $\beta_i' := \beta_i|_{[0,t_i^*]}$ and $z_i' := \beta_i(t_i^*) \in G_0$, so that $\beta_i' \cap G_0 = \{z_i'\}$.

Assemble the certificate. Define the rerouted port arcs $\tilde{\alpha}_i := \alpha_i * \beta_i'$ for $1 \le i \le s$, the concatenation of α_i and β_i', reparametrized over $[0,1]$. Each $\tilde{\alpha}_i$ is a Jordan arc: both pieces are Jordan arcs meeting exactly at $y_i = \alpha_i(1) = \beta_i'(0)$, because $\beta_i' \setminus \{y_i\} \subset \Omega'$ while the image $\alpha_k([0,1])$ of every port lies in $Q \cup \bigcup_j \alpha_j\big((0,1]\big)$, a set disjoint from Ω'. The same disjointness, together with the pairwise disjoint balls containing the β_i, shows that $\tilde{\alpha}_1, \dots, \tilde{\alpha}_s$ are pairwise disjoint. Each $\tilde{\alpha}_i$ meets A_i exactly in $\{x_i\}$, and its interior satisfies $\tilde{\alpha}_i\big((0,1]\big) \subset \Omega_0 \subset \Omega_{\mathrm{free}}$: it avoids every atom, every relative terminal protected collar tail, the obstacle and critical-value set, and the localization-forbidden set — exactly the avoidance clauses that Definition 5.1 imposes on free-port arcs. Next, subdivide G_0 so that each z_i' is a vertex, and call the resulting embedded finite tree G. Then $G \subset \Omega'$ avoids the same forbidden sets, joins the outer port endpoints $z_1', \dots, z_s'$, and meets each port arc $\tilde{\alpha}_i$ exactly in its outer endpoint z_i', by the first-hit property and $G \subset \Omega'$. Consequently $G \cup \bigcup_i \tilde{\alpha}_i$ is again a finite tree: it is connected, and attaching finitely many pairwise disjoint arcs to a tree, each along a single endpoint, adds one vertex and one edge at a time and so preserves the tree characterization $(\text{vertices}) - (\text{edges}) = 1$ for connected finite graphs. The planar cyclic order of half-edges at every vertex is the one induced by the embedding, and it is retained as the certificate's ribbon datum. Finally, the protected prime-end crosscut chains, with their pairwise disjoint relative terminal collar tails, belong to the frozen inventory and are avoided as recorded above. All clauses of Definition 5.1 are satisfied: the rerouted ports $\tilde{\alpha}_i$ together with the spanning tree G constitute a protected ribbon-routing certificate. Theorem 5.4 now realizes it by a compact connected full carrier

with singleton attachment traces, protected crosscut survival, and preserved cyclic order.

Persistence. For the final assertion, suppose that no component of $U \setminus Q$ has simultaneous singleton ports to every required atom. Let a proposed connector be enlarged in any way that retains the same Q and continues to satisfy the certificate's avoidance requirements. If the enlarged data formed a certificate, the necessity argument would place its connected routing part in one component of $U \setminus Q$ and would make its own port arcs simultaneous singleton ports from that component, contrary to the hypothesis. Thus no such enlargement can become a certificate. Changing this conclusion requires changing the frozen inventory itself, which is outside the asserted operation. □

The construction is natural under univalent changes of the Jordan localization. An orientation-preserving conformal homeomorphism of Jordan disks extends over the Jordan boundaries, and it carries the frozen set Q, the free domain (19) and its components, the port arcs, the escape arcs, and the spanning tree to their transported counterparts; the transported data form a certificate for the transported inventory, and the realized carriers correspond as in Proposition 5.6. At the same time the criterion is qualitative: the number of ports and of ribbon sides is not bounded uniformly, and no width, modulus, or incidence estimate enters at any point. Nothing in this section constructs the frozen inventory of a dynamical situation; the corresponding virtual free-component questions are stated in the source manuscript only as open problems [12], and no part of this paper bears on them.

7.4. **A separator between individually accessible atoms.** The criterion of Theorem 7.3 is strictly stronger than every atom-by-atom access condition. The following example, promised in the introduction, exhibits two atoms each of which passes every individual-access audit, while no common free component exists; it is the minimal model of the failure mode that the persistence clause certifies. See Figure 6.

Example 7.4 (Individual accesses without a common free component)**.** Let $U := \{z \in \mathbb{C} : |z| < 2\}$ and take the two atoms

$$A_1 := \left[-\tfrac{1}{4}, \tfrac{1}{4}\right], \qquad A_2 := \left[\tfrac{5}{4}, \tfrac{7}{4}\right],$$

closed segments of the real axis, each a full continuum compactly contained in U. Freeze exactly one further inventory item, the round circle

$$C := \{z : |z| = 1\},$$

recorded as a separated obstacle; all that matters below is that $C \subset Q$, so any recording of C in the frozen inventory would do. There are no critical values and no further forbidden data, so the frozen set is $Q = A_1 \cup A_2 \cup C$ and

$$\Omega_{\text{free}} = U \setminus (A_1 \cup A_2 \cup C).$$

The free domain has exactly two components, one per atom. Write

$$\Omega_{\text{in}} := \{|z| < 1\} \setminus A_1, \qquad \Omega_{\text{ann}} := \{1 < |z| < 2\} \setminus A_2,$$

so that $\Omega_{\text{free}} = \Omega_{\text{in}} \sqcup \Omega_{\text{ann}}$, a disjoint union of open sets separated by the circle C. Each is connected, by an explicit radial–angular path argument. In Ω_{in}, every point can be joined to the circle $\{|z| = \frac{1}{2}\}$, which misses A_1: a point with $\arg z \notin \{0, \pi\}$ travels radially to modulus $\frac{1}{2}$, and the radial path keeps its argument, hence stays off the real axis and misses the real segment A_1; a real point of Ω_{in} has modulus in $(\frac{1}{4}, 1)$ and travels radially to modulus $\frac{1}{2}$ through real points of modulus $> \frac{1}{4}$, which also miss A_1. In Ω_{ann}, every point can be joined to the circle $\{|z| = \frac{9}{8}\}$, which misses A_2: a point with $\arg z \neq 0$ travels radially, inward or outward as needed, keeping its argument; a positive real point of Ω_{ann} has modulus in $(1, \frac{5}{4}) \cup (\frac{7}{4}, 2)$, and in the first case it travels radially to modulus $\frac{9}{8}$ through positive real points of modulus $< \frac{5}{4}$, while in the second case it first travels along its own circle $\{|z| = \text{const}\}$ (which misses A_2, as $\frac{7}{4} < |z| < 2$) to argument π and then proceeds radially inward as before. Both sets are therefore domains, and they are exactly the two components of Ω_{free}.

Every individual-access audit passes. Exhibit the singleton ports

$$\alpha_1(t) := \tfrac{1}{4} + \tfrac{t}{2}, \qquad \alpha_2(t) := \tfrac{7}{4} + \tfrac{t}{8} \qquad (0 \le t \le 1) :$$

the arc α_1 runs from $x_1 = \frac{1}{4} \in A_1$ to $\frac{3}{4}$ and meets A_1 only at x_1, with $\alpha_1\big((0,1]\big) = (\frac{1}{4}, \frac{3}{4}] \subset \Omega_{\text{in}}$; the arc α_2 runs from $x_2 = \frac{7}{4} \in A_2$ to $\frac{15}{8}$ and meets A_2 only at x_2, with $\alpha_2\big((0,1]\big) = (\frac{7}{4}, \frac{15}{8}] \subset \Omega_{\text{ann}}$. Thus each atom has a singleton port into *its own* free component. In the language of the port audit, with empty protected marking and with the atoms as their own completed packets, both unprotected exposure loci are nonempty, so the hypotheses of Lemma 9.2 hold and its conclusion is correct: the two arcs can be (and are) chosen pairwise disjoint with interiors in the common complement of the packets. Atom by atom, nothing is wrong.

No common free component exists. Every port of A_1 runs in Ω_{in}: if α is an arc with $\alpha \cap A_1 = \{\alpha(0)\}$ and $\alpha((0,1])$ contained in one component of Ω_{free}, then $\alpha(t) \to \alpha(0) \in A_1 \subset \{|z| \le \frac{1}{4}\}$ forces $|\alpha(t)| < 1$ for small t, so the component is Ω_{in}. Likewise every port of A_2 runs in Ω_{ann}, since points of Ω_{free} near A_2 have modulus in $(1, 2)$. As $\Omega_{\text{in}} \neq \Omega_{\text{ann}}$, no component of Ω_{free} has simultaneous singleton ports to both atoms, and Theorem 7.3 yields: *no protected ribbon-routing certificate exists* for this frozen inventory. By the theorem's final assertion, no connector enlargement that continues to respect this fixed Q can become a certificate. The only cure would be to delete C from the frozen inventory — a revision of the frozen data, which the formalism forbids.

The example separates two layers of the theory cleanly. The hypotheses of Lemma 9.2 — individual exposure of every atom — are statements about each atom in isolation, and they hold here; the criterion of Theorem 7.3 is

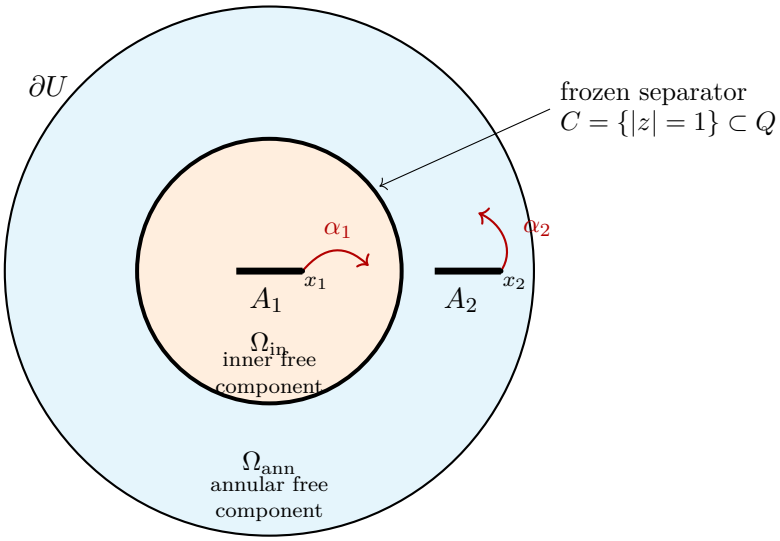


FIGURE 6. The separation example of Example 7.4. The Jordan localization is the disk $U = \{|z| < 2\}$; the frozen inventory consists of the two atoms $A_1 = [-\frac{1}{4}, \frac{1}{4}]$ and $A_2 = [\frac{5}{4}, \frac{7}{4}]$ and the bold circle C, a frozen separator belonging to Q. The free domain $\Omega_{\text{free}} = U \setminus Q$ has exactly two components, shaded in different tints: the inner component Ω_{in} and the annular component Ω_{ann}. Each atom has a singleton port arc (α_1, α_2) into its own component, so every individual-access audit passes; but no single component ports to both atoms, so by Theorem 7.3 no protected ribbon-routing certificate exists, and by the persistence clause no monotone connector enlargement inside U can repair the failure while C stays in the frozen inventory.

a statement about one common component, and it fails. This is also precisely why Definition 8.1 demands that the completed packets be pairwise disjoint *and cover* Q: the separator C must be assigned to some packet. Assigning C to a separated-obstacle packet gives $\text{Fill}(C) = \{|z| \le 1\} \supset A_1$, a packet merger with the completed packet of A_1, and this merger is persistent by Theorem 9.3. Assigning C directly to either atom label makes the corresponding raw seed disconnected, so clause (1) of the packet definition fails unless additional declared connecting material is included in the frozen packet input. Adding such material changes the input and requires a fresh audit; it is not a repair of the fixed assignment. Thus the packet formalism detects the obstruction consistently with the exact criterion, although only the obstacle-label case invokes the theorem's monotone-persistence conclusion.

8. Augmented atoms and the common domain

The free-component criterion of Theorem 7.3 is exact, and its persistence clause makes failures permanent; but as a *positive* test it is expensive. Deciding whether one component of Ω_{free} reaches every required atom demands a global analysis of how each of the finitely many collars, obstacles, and exclusions of the frozen inventory sits relative to all of the others, and Example 7.4 shows that no atom-by-atom inspection can substitute for that analysis: individual accessibility of every atom is compatible with an unrepairable separation. From a *partial* package — the situation in which the criterion would actually be invoked, with the inventory delivered piecemeal by an ambient proof program — the exact free-domain test is therefore difficult to verify directly [12].

The present section proves a sufficient criterion that is *source-checkable*: each of its clauses is verified against named, finite source data rather than against the component structure of an open set. The idea is to reverse the direction of the analysis. Instead of asking how the frozen set Q decomposes the localization, we package Q itself into finitely many pairwise disjoint full continua — each atom augmented by the inventory pieces assigned to it, the remainder grouped into separated-obstacle packets — while reserving one free port for each original atom. Once the completed packets are pairwise disjoint, sphere duality forces their common complement to be *connected*; connectedness, not component bookkeeping, then delivers the common free component that Theorem 7.3 demands.

The section has three parts. Definition 8.1 fixes the packet structure and its typed assignment record; Lemma 8.3 is the duality argument; and Theorem 8.4 assembles the sufficient criterion. How the completed packets are actually computed from raw assigned pieces, and how their defining clauses are audited with persistent failure certificates, is the subject of Section 9; Example 9.4 there shows why the audit must be run on the completions and not on the raw pieces.

8.1. **Filled hulls and augmented packets.** Recall from Section 2 the filled hull of a compact plane set K:

$$\mathrm{Fill}(K) \;=\; K \,\cup\, \bigcup\big\{\, B : B \text{ a bounded component of } \mathbb{C}\setminus K \,\big\} \tag{20}$$

(equation (9.5) of [12]). Since K is compact, $\mathbb{C}\setminus K$ has exactly one unbounded component $\Omega_\infty(K)$, and (20) says

$$\mathrm{Fill}(K) \;=\; \mathbb{C}\setminus \Omega_\infty(K).$$

Consequently $\mathrm{Fill}(K)$ is closed, and it is bounded because every closed round disk containing K has its exterior inside $\Omega_\infty(K)$; thus $\mathrm{Fill}(K)$ is compact. Its complement is the single connected set $\Omega_\infty(K)$, so $\mathrm{Fill}(K)$ is *full*, and it is the smallest full compact set containing K: if $F \supset K$ is full and compact, then $\mathbb{C}\setminus F$ is a connected unbounded subset of $\mathbb{C}\setminus K$, hence contained in $\Omega_\infty(K)$, which gives $\mathrm{Fill}(K) \subset F$ after taking complements.

Here neither K nor its filled hull is asserted to be connected; this least-full-compact statement does not produce the least full *continuum* ruled out by Theorem 3.1. Finally, if K is a continuum, then so is $\mathrm{Fill}(K)$: every bounded component B of $\mathbb{C}\setminus K$ has $\partial B \subset K$, so its closure $\overline{B}$ is a connected set meeting K, and $\mathrm{Fill}(K) = K \cup \bigcup \overline{B}$ is a union of connected sets all meeting the connected set K. The passage from K to $\mathrm{Fill}(K)$ is what the manuscript calls *fullification*; its algebra — and in particular the fact that interleaved adjoin-and-fill operations compute a well-defined completion — is developed in Section 9.

Throughout, $s \geq 1$, and the *atoms* $A_1, \dots, A_s \Subset U$ are the pairwise disjoint full continua of Section 5 that the connector must join, and Q is the frozen set of Section 7 (equation (9.4) of [12]): the union of the atoms, the relative terminal protected collar tails, the separated obstacles, the critical-value exclusion, and the localization-forbidden set, fixed once and for all before any connector is chosen. Retain the compact atom–tail blocks $\mathcal{B}_i$ of Section 7.1, and choose a finite named compact decomposition $Z = Z_1 \cup \cdots \cup Z_N$ of the remaining frozen set (the one-piece decomposition is allowed; if $Z = \varnothing$, take $N = 0$ and no Z_r). The packet input pieces are the $\mathcal{B}_i$ and the Z_r; a bare relative collar tail, which need not be compact, is never processed separately.

Definition 8.1 (Augmented-atom packet; manuscript Definition 10.9)**.** For the frozen set Q, an *augmented-atom packet* consists of full continua

$$\widehat{A}_1, \dots, \widehat{A}_s,\ \widehat{O}_1, \dots, \widehat{O}_t \ \Subset\ U$$

and pairwise disjoint port arcs $\alpha_i \colon [0,1] \to U$ such that:

(1) each atom–tail block $\mathcal{B}_i$ is assigned to its own atom label, every named compact constituent Z_r has exactly one assignment to an atom or separated-obstacle label, and every resulting atom seed $\mathcal{B}_i \cup E_i$ and obstacle seed O_ℓ is nonempty, compact, and connected;
(2) $\widehat{A}_i = \mathrm{Fill}(\mathcal{B}_i \cup E_i)$ for every i, and $\widehat{O}_\ell = \mathrm{Fill}(O_\ell)$ for every ℓ;
(3) the completed packets $\widehat{A}_1, \dots, \widehat{A}_s, \widehat{O}_1, \dots, \widehat{O}_t$ are pairwise disjoint and cover Q; and
(4) $\alpha_i(0) = x_i \in A_i$, $\alpha_i \cap \widehat{A}_i = \{x_i\}$, and

$$\alpha_i\big((0,1]\big) \ \subset\ U \setminus \Big(\bigcup_k \widehat{A}_k \cup \bigcup_\ell \widehat{O}_\ell\Big),$$

and x_i belongs to no impression of a protected chain.

The *typed assignment* is the resulting function from the finitely many input pieces $\mathcal{B}_i, Z_r$ to the atom and separated-obstacle packet labels, with each $\mathcal{B}_i$ fixed at label i.

Some commentary on the four clauses. In clause (1), E_i denotes the union of the compact Z_r assigned to the i-th atom; it may be empty, and the clause demands that the resulting *atom seed* $\mathcal{B}_i \cup E_i$ be compact and connected, so that an assigned piece disjoint from its block must be accompanied by

connecting material assigned to the same label. The assignment is load-bearing: it is what makes a later change of packet label an explicit, auditable re-decision rather than a silent repair, a point Example 9.4 makes concrete.

In clause (2), fullness of the completed packets is automatic: the seeds are compact and connected, so their filled hulls are full continua by the discussion following (20). Clause (3) is the substantive geometric hypothesis, and it is stated for the *completed* packets $\widehat{A}_i, \widehat{O}_\ell$, not for the raw seeds: fullification can create intersections that the seeds do not have, and an audit run at seed level certifies nothing (Example 9.4). Coverage of Q means that every item of the frozen inventory lies inside some completed packet, so that outside the packets there is no frozen material at all — this is the inclusion that will place the common complement inside Ω_{free}. Clause (4) is the port clause. It is deliberately stronger than the port clause of Definition 5.1: the arc's positive-parameter part must avoid every completed hull, whereas the arcs of Definition 5.1 avoid only the obstacle set, the atoms, and the relative terminal collar tails, and the arc here must meet even its own augmented atom $\widehat{A}_i$ only at the landing point x_i. The final requirement, that x_i belong to no impression of a protected chain, keeps the port compatible with the protected marking; it reappears in Section 9 as the requirement that x_i lie in the unprotected exposure locus, outside the forbidden impression set P_i.

8.2. **Nonseparation of finite disjoint full families.** The engine of the common-domain theorem is a nonseparation statement for finitely many disjoint full continua on the sphere. Here a compact set $K \subset S^2$ is called full when $S^2 \setminus K$ is connected; for a compact subset of the plane this agrees with the planar notion, because adjoining the point at infinity to the unbounded complementary component does not change connectivity, so a plane-full compact set is sphere-full and conversely. The proof uses the compacta form of Alexander duality [15, Chapter 6]:

$$\widetilde{H}_0\big(S^2 \setminus K; \mathbb{Z}\big) \;\cong\; \check{H}^1(K; \mathbb{Z}) \tag{21}$$

for an arbitrary nonempty proper compact subset $K \subset S^2$ (equation (9.6) of [12]). The Čech form, rather than the locally contractible special case, is required here: the atoms and the raw packet seeds are arbitrary continua, no local connectivity is assumed anywhere in this paper, and for such compacta singular cohomology need not compute the group on the right of (21).

Remark 8.2. The insistence on the Čech form is not pedantry but a structural feature of the framework. Clause (6) of Theorem 5.4 promises that no local connectivity of any atom is ever required, and the packets of Definition 8.1 inherit whatever boundary pathology their atoms and assigned pieces possess. Čech cohomology is the invariant that remains dual to the complement's component count for every compact subset of the sphere, without any tameness hypothesis; every duality argument below is stated so that it factors through (21) alone.

Lemma 8.3 (Finite disjoint full packets; manuscript Lemma 10.10)**.** *If $K_0, \dots, K_N \subsetneq S^2$ are pairwise disjoint nonempty full continua, then*

$$S^2 \setminus \bigcup_{r=0}^{N} K_r$$

is connected.

Proof. Write $X := \bigcup_{r=0}^{N} K_r$. The set X is a proper subset of S^2, so that (21) applies to X and to each K_r: were $X = S^2$, the sets $K_0, \dots, K_N$ would be finitely many pairwise disjoint nonempty closed sets covering S^2; each would then be open as well, being the complement of the union of the others, and connectedness of S^2 would force $N = 0$ and $K_0 = S^2$, in contradiction with $K_0 \subsetneq S^2$.

Step 1: each packet has vanishing first Čech cohomology. Fix r. Fullness of K_r makes $S^2 \setminus K_r$ connected, and it is nonempty because K_r is proper. Being open in S^2, the complement is locally path connected, so connectedness makes it path connected and its reduced zeroth singular homology vanishes. Hence (21), applied to the nonempty proper compact set K_r, gives

$$\check{H}^1(K_r; \mathbb{Z}) \;\cong\; \widetilde{H}_0\big(S^2 \setminus K_r; \mathbb{Z}\big) \;=\; 0.$$

Step 2: disjoint neighborhoods split fine covers. If $N = 0$, the conclusion is exactly the assumed fullness of K_0, so assume $N \geq 1$. The sets $K_0, \dots, K_N$ are finitely many pairwise disjoint compacta, so their mutual distances in the spherical metric are positive; choose $\varepsilon > 0$ smaller than half the minimum of these distances and let N_r be the open ε-neighborhood of K_r. The N_r are pairwise disjoint open sets with $K_r \subset N_r$.

Call a cover of X by relatively open subsets *fine* when every member is contained in one of the sets N_r. The fine covers are cofinal among all relatively open covers of X: an arbitrary cover $\mathcal{U}$ is refined by the fine cover $\{\, V \cap N_r : V \in \mathcal{U},\ 0 \leq r \leq N \,\}$, since $X \subset \bigcup_r N_r$. Let $\mathcal{V}$ be a fine cover with nonempty members. Each member lies in exactly one N_r, because the N_r are pairwise disjoint; write $\mathcal{V}_r$ for the members contained in N_r. Since $X \cap N_r = K_r$, the collection $\mathcal{V}_r$ is a relatively open cover of K_r. Two members taken from distinct collections $\mathcal{V}_r, \mathcal{V}_{r'}$ are disjoint, lying in the disjoint sets $N_r, N_{r'}$; hence no simplex of the nerve of $\mathcal{V}$ mixes the collections, and the nerve of $\mathcal{V}$ is the disjoint union of the nerves of $\mathcal{V}_0, \dots, \mathcal{V}_N$. Simplicial cohomology of a finite disjoint union is the direct sum, so

$$H^1\big(\operatorname{nerve}\mathcal{V}; \mathbb{Z}\big) \;\cong\; \bigoplus_{r=0}^{N} H^1\big(\operatorname{nerve}\mathcal{V}_r; \mathbb{Z}\big).$$

Step 3: pass to the direct limit. Čech cohomology of X is the direct limit of the nerve cohomologies over the directed system of relatively open covers, and the fine covers are cofinal in that system, so the limit may be computed over fine covers only. A refinement of a fine cover is again fine after intersecting with the N_r, and its restriction to N_r refines $\mathcal{V}_r$; the splitting

of Step 2 is therefore compatible with the refinement maps, and restriction to N_r is cofinal among the covers of K_r because any relatively open cover of K_r extends to a fine cover of X by adjoining covers of the other packets. Since a direct limit commutes with the finite direct sum, Step 1 gives

$$\check{H}^1(X;\mathbb{Z}) \;\cong\; \bigoplus_{r=0}^{N} \check{H}^1(K_r;\mathbb{Z}) \;=\; 0.$$

Step 4: apply duality a second time. The set X is compact, nonempty, and proper, so (21) applies to X itself:

$$\widetilde{H}_0\big(S^2 \setminus X;\mathbb{Z}\big) \;\cong\; \check{H}^1(X;\mathbb{Z}) \;=\; 0.$$

The complement $S^2 \setminus X$ is nonempty. Vanishing reduced singular zeroth homology makes it path connected (and therefore connected); equivalently, one may note that this open subset of the sphere is locally path connected, so its connected and path components agree. This is the assertion. □

Lemma 8.3 is the disjoint counterpart of Lemma 5.2. There, unions of nonseparating continua meeting in connected singletons were handled by the Janiszewski separation lemma, one attachment at a time; here the continua are pairwise disjoint, the count is arbitrary, and the duality argument dispenses with any hypothesis on how the packets face one another. Nothing is assumed about the boundaries of the K_r — they may be nowhere locally connected — and no width, diameter, or separation estimate enters.

8.3. **The common-domain theorem.**

Theorem 8.4 (Augmented-Atom Common-Domain Theorem; manuscript Theorem 10.11)**.** *Every augmented-atom packet determines one component of Ω_{free} with simultaneous singleton ports to every original atom. Hence it supplies a protected ribbon-routing certificate and its realized carrier.*

Proof. Let an augmented-atom packet be given, with completed packets $\widehat{A}_1,\dots,\widehat{A}_s,\widehat{O}_1,\dots,\widehat{O}_t \Subset U$ and port arcs $\alpha_1,\dots,\alpha_s$ as in Definition 8.1.

The exterior packet. Adjoin to the family the set

$$K_0 \;:=\; S^2 \setminus U.$$

The Jordan-disk hypothesis makes K_0 a full continuum: by the Jordan–Schoenflies theorem the Jordan curve ∂U divides the sphere into two open Jordan disks with common boundary ∂U, the closure of the one containing ∞ is the closed Jordan disk $S^2 \setminus U$, hence a continuum, and its complement is the connected set U, so K_0 is full in the sphere sense used in Lemma 8.3. Each completed packet is a plane-full continuum, hence sphere-full as noted in Section 8.2, and compact containment $\widehat{A}_i,\widehat{O}_\ell \Subset U$ makes every completed packet disjoint from K_0; together with clause (3) of Definition 8.1, the family

$$K_0,\ \widehat{A}_1,\dots,\widehat{A}_s,\ \widehat{O}_1,\dots,\widehat{O}_t$$

consists of $1+s+t$ pairwise disjoint nonempty full continua in S^2.

The augmented complement is connected. Lemma 8.3 applies to this family and makes its common complement connected. Since $S^2 \setminus K_0 = U$, that complement is

$$\Omega_{\mathrm{aug}} \;:=\; U \setminus \Big(\bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big) \;=\; S^2 \setminus \Big(K_0 \cup \bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big), \tag{22}$$

a connected subset of $\mathbb{C}$, open because U is open and the packets are compact. It is nonempty: clause (4) places the positive-parameter part of each port arc inside it.

The augmented complement lies in one free component. Clause (3) of Definition 8.1 makes the completed packets cover Q, so

$$\Omega_{\mathrm{aug}} \;=\; U \setminus \Big(\bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big) \;\subset\; U \setminus Q \;=\; \Omega_{\mathrm{free}}.$$

A nonempty connected subset of the open set Ω_{free} lies in exactly one of its components; let Ω_0 be the component containing Ω_{aug}. This is the component that the packet determines.

The ports are simultaneous singleton ports of Ω_0. By clause (4), each port interior satisfies

$$\alpha_i\big((0,1]\big) \;\subset\; U \setminus \Big(\bigcup_k \widehat{A}_k \cup \bigcup_\ell \widehat{O}_\ell\Big) \;=\; \Omega_{\mathrm{aug}} \;\subset\; \Omega_0.$$

Moreover $A_i \subset \mathcal{B}_i \cup E_i \subset \widehat{A}_i$, so

$$\alpha_i \cap A_i \;\subset\; \alpha_i \cap \widehat{A}_i \;=\; \{x_i\},$$

and $x_i = \alpha_i(0) \in A_i$ gives $\alpha_i \cap A_i = \{x_i\}$: the arc meets its atom exactly in the one free point x_i. The arcs are pairwise disjoint Jordan arcs by Definition 8.1. Thus Ω_0 has simultaneous singleton ports to every original atom, in the sense of Section 7.

Conclusion. Theorem 7.3 converts the simultaneous singleton ports of Ω_0 into a protected ribbon-routing certificate, and Theorem 5.4 realizes that certificate by a compact connected full carrier with singleton attachment traces, protected crosscut survival, and preserved cyclic order. The packet therefore supplies a protected ribbon-routing certificate and its realized carrier. □

Remark 8.5. Three remarks of the manuscript [12] deserve emphasis. First, no local connectivity or width estimate occurs anywhere in the argument: the packets may have arbitrarily wild boundaries, the duality step of Lemma 8.3 is insensitive to them, and no modulus, diameter, or incidence bound is invoked at any point. Second, the construction is natural under an orientation-preserving conformal homeomorphism of Jordan disks: the boundary extensions can be glued to a sphere homeomorphism, and such a map carries filled packets to filled packets, ports to ports, the common domain (22) to the corresponding common domain, and any selected connector to a connector for the transported data, exactly as in Proposition 5.6;

the commutation of such a sphere homeomorphism with the operator Fill is proved in detail in the naturality paragraph of Theorem 9.3. Third, the spanning tree produced inside the common component is not claimed canonical: Theorem 7.3 supplies *some* embedded tree through the free component, and different choices give different carriers with the same protected marking. Only the component Ω_0, the ports, and the survival data are determined by the packet.

Theorem 8.4 consumes completed packets: it presupposes that the filled hulls $\widehat{A}_i, \widehat{O}_\ell$ have already been computed from the typed assignment and checked against clauses (1)–(4). In practice the assignment delivers raw pieces one at a time, and both the computation of the completions and the verification of the clauses raise questions of their own — whether the result depends on the order of processing, and whether individually verified ports must be untangled globally. The next section resolves both questions and equips the verification with persistent failure certificates.

9. Fixed-packet closure and the port audit

Theorem 8.4 consumes completed packets, and its hypotheses leave two apparent degrees of freedom in producing them. The first is a processing order. The typed assignment of Definition 8.1 delivers each packet as a finite list of compact pieces, and a completion built incrementally — adjoin a piece, fill, adjoin the next piece carrying the same label, fill again — seems to depend on the order in which the pieces arrive. The second is a global untangling. Clause (4) demands pairwise disjoint port arcs avoiding every completed hull simultaneously, whereas what an ambient proof program can realistically certify is one access arc per atom, each verified in isolation; one might fear that the individual arcs must be coordinated globally before they can serve together. This section proves that both dependencies are illusory once the typed assignment is fixed. Fullification is order-independent (Lemma 9.1); individual ports localize simultaneously without any coordination (Lemma 9.2); and the resulting audit (Theorem 9.3) characterizes the geometric packet clauses by three checkable conditions whose failures are certified by witnesses that *persist* under every monotone same-label enlargement. Example 9.4, required by the structure of the theory, then shows that the audit must be run on the completions and not on the raw seeds, and Proposition 9.5 packages the whole verification into a cellwise checklist.

9.1. **Order-independent fullification.**

Lemma 9.1 (Filled-hull absorption; manuscript Lemma 10.12)**.** *For compact plane sets K, L,*

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) \;=\; \mathrm{Fill}(K \cup L) \tag{23}$$

(equation (9.7) of [12]*). Consequently, sequentially adjoining and filling finitely many compact pieces gives the filled hull of their union, independently of processing order.*

Proof. Step 1: the operator Fill *is extensive, monotone, and idempotent.* Extensivity, $K \subset \mathrm{Fill}(K)$, is immediate from the definition (20). For monotonicity, recall from Section 8.1 that

$$\mathrm{Fill}(K) \;=\; \mathbb{C} \setminus \Omega_\infty(K),$$

where $\Omega_\infty(K)$ is the unique unbounded component of $\mathbb{C}\setminus K$, and that $\mathrm{Fill}(K)$ is compact. If $K \subset L$, then $\Omega_\infty(L)$ is a connected unbounded subset of $\mathbb{C} \setminus L \subset \mathbb{C} \setminus K$, hence is contained in the unique unbounded component of $\mathbb{C} \setminus K$:

$$\Omega_\infty(L) \;\subset\; \Omega_\infty(K).$$

Taking complements in $\mathbb{C}$ gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$. For idempotence, the complement of $\mathrm{Fill}(K)$ is the single set $\Omega_\infty(K)$, which is connected and unbounded; a compact set whose complement has exactly one component, an unbounded one, has no bounded complementary components, so $\mathrm{Fill}(\mathrm{Fill}(K)) = \mathrm{Fill}(K)$. We call a compact set of the form $\mathrm{Fill}(M)$, equivalently a compact set equal to its own filled hull, *full*; by idempotence, $\mathrm{Fill}(K \cup L)$ is full.

Step 2: the two inclusions. Extensivity applied to the summand K gives $K \cup L \subset \mathrm{Fill}(K) \cup L$, and monotonicity then gives the first inclusion,

$$\mathrm{Fill}(K \cup L) \;\subset\; \mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big).$$

Conversely, monotonicity gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(K \cup L)$, and extensivity gives $L \subset K \cup L \subset \mathrm{Fill}(K \cup L)$; hence

$$\mathrm{Fill}(K) \cup L \;\subset\; \mathrm{Fill}(K \cup L),$$

and the set on the right is already full by Step 1. Applying Fill to this second inclusion and using monotonicity and idempotence,

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) \;\subset\; \mathrm{Fill}\big(\mathrm{Fill}(K \cup L)\big) \;=\; \mathrm{Fill}(K \cup L).$$

The two displayed inclusions prove (23).

Step 3: finite order independence. Let $P_1, \dots, P_n$ be compact pieces and let π be any permutation of $\{1, \dots, n\}$. Define the incremental completions

$$X_1 := \mathrm{Fill}\big(P_{\pi(1)}\big), \qquad X_j := \mathrm{Fill}\big(X_{j-1} \cup P_{\pi(j)}\big) \quad (2 \le j \le n).$$

We claim $X_j = \mathrm{Fill}\big(P_{\pi(1)} \cup \cdots \cup P_{\pi(j)}\big)$ for every j, by induction. The base case is the definition of X_1. For the step, write $K := P_{\pi(1)} \cup \cdots \cup P_{\pi(j-1)}$ and $L := P_{\pi(j)}$; the inductive hypothesis and (23) give

$$X_j = \mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) = \mathrm{Fill}(K \cup L) = \mathrm{Fill}\big(P_{\pi(1)} \cup \cdots \cup P_{\pi(j)}\big).$$

In particular $X_n = \mathrm{Fill}(P_1 \cup \cdots \cup P_n)$, which does not depend on π. The same identity shows that filling may be interleaved arbitrarily, and even that two partially completed runs of the same label may be merged: by (23) applied twice, once on each side of the union,

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup \mathrm{Fill}(L)\big) = \mathrm{Fill}\big(K \cup \mathrm{Fill}(L)\big) = \mathrm{Fill}(K \cup L)$$

for all compact K, L. Every schedule of adjoin-and-fill operations on a fixed finite set of pieces therefore computes the same completion. □

9.2. **Seeds, completions, and persistent witnesses.** For a fixed packet label λ, let S_λ be the union of its assigned compact input pieces: for an atom label i this union contains the mandatory block $\mathcal{B}_i$, and for a separated-obstacle label it contains only assigned Z_r. This nonempty set is the *raw seed* of the packet. Its unique completion is

$$K_\lambda \;=\; \mathrm{Fill}(S_\lambda) \tag{24}$$

(equation (9.8) of [12]); by Lemma 9.1 this is what every processing order computes. Note that when the label carries an atom, the convention makes $A_i \subset \mathcal{B}_i \subset S_i \subset K_i$. When atom and obstacle labels must be distinguished, we write K_i for the former completions and L_ℓ for the latter.

Failures of the packet clauses are recorded by explicit geometric witnesses. For $x \in K_\ell$, record its *chamber*,

$$\mathrm{ch}_\ell(x) \;:=\; \begin{cases} \mathsf{seed} & \text{if } x \in S_\ell, \\ \text{the bounded component of } \mathbb{C} \setminus S_\ell \text{ containing } x & \text{otherwise.} \end{cases}$$

A completed-packet merger then has the geometric witness

$$\big(\ell,\ \ell',\ x,\ \mathrm{ch}_\ell(x),\ \mathrm{ch}_{\ell'}(x)\big), \qquad x \in K_\ell \cap K_{\ell'} \tag{25}$$

(display (9.9) of [12]): the tuple names the two colliding labels, a common point, and, for each label, whether the point is seed material or lies in a named bounded complementary chamber of the seed. Monotonicity of Fill shows that this witness remains a merger after any same-label enlargement $S_\ell \subset S'_\ell$: enlarging seeds enlarges completions, so $x \in K_\ell \cap K_{\ell'} \subset K'_\ell \cap K'_{\ell'}$, and the same point witnesses the merger of the enlarged packets. (The recorded chambers may upgrade — a bounded chamber of S_ℓ may be absorbed into the seed of S'_ℓ, in which case the record becomes seed — but the intersection never empties.) The same argument makes a *localization escape*

$$K_\ell \cap (\mathbb{C} \setminus U) \;\neq\; \varnothing$$

persistent: a point of K_ℓ outside U lies in K'_ℓ outside U for every same-label enlargement.

The third audited datum concerns ports. Let $P_i \subset A_i$ be the union of the impressions forbidden by the protected marking at the last proof use; the protected marking forbids exactly the impressions of the protected chains, so P_i is the union of those impressions inside A_i, and we use this identification in both directions of the audit below. Define the *unprotected exposure locus*

$$\begin{aligned} \mathrm{Exp}(A_i, K_i; P_i) \;=\; \big\{\, x \in A_i \setminus P_i \;:\; &\text{there is a Jordan arc } \gamma\colon [0,1] \to U \\ &\text{with } \gamma(0) = x \text{ and } \gamma \cap K_i = \{x\} \,\big\} \end{aligned} \tag{26}$$

(display (9.10) of [12]). This is actual access data; boundary membership alone is not substituted for (26). The distinction matters precisely because

no local connectivity is assumed: a point of $\partial K_i \setminus P_i$ need not be the landing point of any arc — the boundary of a full continuum can consist almost entirely of non-accessible points [14] — so (26) is an existence statement about arcs, and it is this statement, not membership in a boundary set, that every clause below reads.

9.3. **Simultaneous localization of individual ports.**

Lemma 9.2 (Individual ports localize simultaneously; manuscript Lemma 10.13)**.** *Let $s \geq 1$. Suppose the completed atom packets $K_1, \dots, K_s \Subset U$ and obstacle packets $L_1, \dots, L_t \Subset U$ form a pairwise disjoint family and $\mathrm{Exp}(A_i, K_i; P_i) \neq \varnothing$ for every original atom. Then the access arcs can be chosen pairwise disjoint, with positive-parameter parts in*

$$U \setminus \Big(\bigcup_i K_i \cup \bigcup_\ell L_\ell \Big).$$

If $K_i \subset K_i'$, then

$$\mathrm{Exp}(A_i, K_i'; P_i) \;\subset\; \mathrm{Exp}(A_i, K_i; P_i) \tag{27}$$

(display (9.11) of [12]*).*

Proof. Disjoint localization neighborhoods. The finite family formed by the K_i and L_ℓ is pairwise disjoint, and every member is compactly contained in the open set U. The finitely many distances from its members to $\mathbb{C} \setminus U$ are positive; when the family has at least two members, so are its finitely many pairwise distances. Choose $\varepsilon > 0$ smaller than half the minimum of the boundary distances together with the pairwise distances when present, and let N_i be the open ε-neighborhood of K_i. Then $N_1, \dots, N_s$ are pairwise disjoint open sets, each compactly contained in U, each containing its K_i, and each missing every obstacle packet L_ℓ and every other atom packet K_j $(j \neq i)$.

Localizing one arc. Fix i and choose, by hypothesis, a point $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$ together with a Jordan arc $\gamma_i \colon [0,1] \to U$ from (26): $\gamma_i(0) = x_i$ and $\gamma_i \cap K_i = \{x_i\}$. The landing point satisfies $x_i \in A_i \subset K_i \subset N_i$, and N_i is open, so continuity of γ_i at 0 supplies $t_i \in (0,1]$ with $\gamma_i([0,t_i]) \subset N_i$. Replace γ_i by its initial subarc $\gamma_i|_{[0,t_i]}$, reparametrized over $[0,1]$ and still denoted γ_i. Restriction can only shrink the intersection with K_i while retaining the landing point, so the restricted arc still satisfies $\gamma_i \cap K_i = \{x_i\}$, and its image now lies in N_i.

Simultaneity for free. The restricted arcs are pairwise disjoint because the N_i are, and no coordination between the choices was needed: each arc was shortened using only its own neighborhood. Each positive-parameter part satisfies

$$\gamma_i\big((0,1]\big) \;\subset\; N_i \setminus K_i,$$

which misses K_i by the singleton-intersection clause and misses every other packet because N_i does; hence it lies in $U \setminus (\bigcup_i K_i \cup \bigcup_\ell L_\ell)$, as asserted.

Monotonicity. Let $K_i \subset K_i'$ and let $x \in \mathrm{Exp}(A_i, K_i'; P_i)$, witnessed by an arc γ with $\gamma(0) = x$ and $\gamma \cap K_i' = \{x\}$. Every arc avoiding the larger packet off its landing point also avoids the smaller one: $\gamma \cap K_i \subset \gamma \cap K_i' = \{x\}$, while $x \in A_i \subset K_i$ keeps x in the intersection, so $\gamma \cap K_i = \{x\}$ and the same arc witnesses $x \in \mathrm{Exp}(A_i, K_i; P_i)$. This proves (27). In particular, an *empty* exposure locus stays empty under every same-label enlargement of the packet: port loss, once incurred, is persistent. □

9.4. **The audit theorem.** By the *geometric augmented-packet clauses* of a fixed finite typed assignment we mean clauses (1)–(4) of Definition 8.1, evaluated on the completions determined by that assignment.

Theorem 9.3 (Fixed-Packet Closure and Port Audit; manuscript Theorem 10.14)**.** *For a fixed finite typed assignment, the geometric augmented-packet clauses hold if and only if:*

(1) *the frozen inventory is covered and every raw seed S_ℓ is nonempty, compact, and connected;*
(2) *the unique completions $K_\ell = \mathrm{Fill}(S_\ell)$ are pairwise disjoint and compactly contained in U; and*
(3) $\mathrm{Exp}(A_i, K_i; P_i) \neq \varnothing$ *for every original atom: the unprotected exposure locus is nonempty.*

The completions are independent of processing order. Merger, localization-escape, and empty-exposure defects persist under monotone same-label enlargement. The complete audit is natural under a conformal homeomorphism of Jordan localizations that transports the entire typed assignment (hence its raw seeds, completed packets, and protected marking).

Proof. Closure. Lemma 9.1 gives unique, order-independent completions $K_\ell = \mathrm{Fill}(S_\ell)$: every schedule of adjoin-and-fill operations on the pieces of a label computes (24). If clause (1) holds, each raw seed is a nonempty compact connected set, so each completion is a full continuum, by the discussion following (20).

Equivalence. Suppose first that clauses (1)–(3) hold. Clause (1) is precisely clause (1) of Definition 8.1: the coverage assertion is common to both, and connectedness of the raw seeds is the connectedness demanded of the atom seeds $\mathcal{B}_i \cup E_i$ and obstacle seeds O_ℓ. Writing $\widehat{A}_i := K_i$ for the atom labels and $\widehat{O}_\ell := L_\ell$ for the obstacle labels, clause (2) of the definition holds by (24), and the sets are full continua by the closure paragraph. Clause (2) of the audit makes these completions pairwise disjoint and, being compact subsets of the open set U, compactly contained in U; coverage of Q follows from clause (1), since $Q \subset \bigcup_\ell S_\ell \subset \bigcup_\ell K_\ell$. This is clause (3) of the definition. Finally, clauses (2) and (3) of the audit satisfy the hypotheses of Lemma 9.2, which produces pairwise disjoint Jordan arcs α_i with $\alpha_i(0) = x_i \in A_i \setminus P_i$, $\alpha_i \cap K_i = \{x_i\}$, and positive-parameter parts in the common complement $U \setminus (\bigcup_k K_k \cup \bigcup_\ell L_\ell)$ — exactly clause (4) of the definition, including its final requirement: P_i contains every impression forbidden by the protected

marking, so $x_i \notin P_i$ means that it belongs to no impression of a protected chain.

Conversely, suppose the geometric packet clauses hold. Clause (1) of the definition is clause (1) of the audit, as before. The completed packets of the definition are the unique completions (24) of the assigned seeds, and clause (3) of the definition makes them pairwise disjoint, while $\widehat{A}_i, \widehat{O}_\ell \Subset U$ gives compact containment: clause (2) of the audit. For clause (3) of the audit, fix an atom A_i. The packet's own port arc α_i satisfies $\alpha_i(0) = x_i \in A_i$, $\alpha_i \cap \widehat{A}_i = \{x_i\}$, and maps into U; since $\widehat{A}_i = K_i$, the arc witnesses $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$ provided $x_i \notin P_i$, and this is the final requirement of clause (4): x_i belongs to no impression of a protected chain, and P_i is the union of those impressions. Hence the exposure locus is nonempty, completing the equivalence.

Persistence. Monotonicity of Fill makes the merger witnesses (25) and the localization-escape witnesses persistent under monotone same-label enlargement, as recorded in Section 9.2: the witnessing point lies in the enlarged completions and outside the unchanged localization, respectively. The monotonicity clause (27) of Lemma 9.2 makes empty exposure persistent: the locus of the enlarged packet is a subset of the old locus, so it cannot become nonempty by enlargement. Thus each of the three defect types, once witnessed, is witnessed forever along every monotone same-label enlargement of the seeds.

Naturality. Let $\varphi\colon U \to V$ be an orientation-preserving conformal homeomorphism of Jordan localizations that transports the full packets and the protected marking. By the Carathéodory extension theorem for Jordan domains [14], φ extends to a homeomorphism $\overline{U} \to \overline{V}$ of the closures; restricting to the boundaries gives a homeomorphism $\partial U \to \partial V$ of Jordan curves. The complementary sets $S^2 \setminus U$ and $S^2 \setminus V$ are closed Jordan disks, and by the Schoenflies theorem [13] the boundary homeomorphism extends to a homeomorphism between them; composing with a self-homeomorphism of the target disk that is the identity on its boundary circle and moves the image of ∞ to ∞, we may arrange that the extension fixes ∞. Gluing along ∂U yields a sphere homeomorphism Φ with $\Phi|_U = \varphi$ and $\Phi(\infty) = \infty$. For any compact $K \subset \mathbb{C}$, the homeomorphism Φ carries the components of $S^2 \setminus K$ bijectively onto the components of $S^2 \setminus \Phi(K)$, and it carries the component containing ∞ to the component containing ∞; deleting ∞, it carries the unbounded complementary component of K to the unbounded complementary component of $\Phi(K)$, and the bounded components to the bounded components. Hence Φ commutes with Fill:

$$\Phi\big(\mathrm{Fill}(K)\big) \;=\; \mathrm{Fill}\big(\Phi(K)\big).$$

Consequently Φ carries raw seeds to raw seeds (piece by piece, with the same packet labels), completions to completions, merger witnesses (25) to merger witnesses (the point x to $\Phi(x)$, the chambers to the corresponding chambers, seed record to seed record), and localization escapes to localization escapes:

$\Phi(\mathbb{C} \setminus U) = \mathbb{C} \setminus V$, so a point of K_ℓ outside U is carried to a point of the transported completion outside V. It also transports the port data: P_i maps to the transported forbidden impression set, and $\gamma \mapsto \Phi \circ \gamma$ is a bijection between the arcs read by (26) in U and those read by the transported locus in V, because $\Phi|_U = \varphi$ is a homeomorphism onto V and $\Phi(\gamma) \cap \Phi(K_i) = \Phi(\gamma \cap K_i)$. Each audit clause therefore holds for the transported assignment if and only if it holds for the original: the complete audit is natural. □

The theorem does not construct the typed assignment. What it proves is that, once the assignment is fixed, a favorable completion order and a global untangling of already existing individual ports are *not* additional degrees of freedom: the completions are canonical by Lemma 9.1, and individual exposure already implies simultaneous localized ports by Lemma 9.2. The one choice that matters — which inventory piece is assigned to which label — is made upstream and recorded in the typed data. The following example, the merger half of the required failure taxonomy, shows that this choice genuinely matters, and that its defects surface only after fullification.

9.5. **A post-fullification packet merger.**

Example 9.4 (Post-fullification packet merger)**.** Work in the Jordan localization $U := \{z \in \mathbb{C} : |z| < 3\}$, and consider a two-packet system whose typed assignment creates the raw seeds

$$S_1 \;:=\; \{z : |z| = 1\}, \qquad S_2 \;:=\; \left[-\tfrac{1}{4}, \tfrac{1}{4}\right] \subset \mathbb{R},$$

say with both labels typed as separated obstacles. Each seed is nonempty, compact, and connected, so clause (1) of Theorem 9.3 holds, and the seeds are pairwise *disjoint*: every point of S_1 has modulus 1, every point of S_2 has modulus at most $\frac{1}{4}$. Consequently every seed-level disjointness audit passes — any inspection of the raw pieces, in any processing order, at any stage of the assignment, sees pairwise disjoint data.

The completions tell a different story. The segment has connected complement, so

$$K_2 \;=\; \mathrm{Fill}(S_2) \;=\; S_2,$$

but the complement of the circle has exactly two components, the bounded open unit disk and the unbounded set $\{|z| > 1\}$, so

$$K_1 \;=\; \mathrm{Fill}(S_1) \;=\; \{z : |z| \le 1\} \;\supset\; S_2.$$

Hence

$$K_1 \cap K_2 \;\supset\; S_2 \;\neq\; \varnothing$$

(indeed $K_1 \cap K_2 = S_2$), and clause (2) of Theorem 9.3 fails: the merger appears only after fullification. The geometric witness (25) at the point $x = 0$ is

$$\big(1,\ 2,\ 0,\ \mathrm{ch}_1(0),\ \mathrm{ch}_2(0)\big), \qquad \mathrm{ch}_1(0) = \{z : |z| < 1\}, \quad \mathrm{ch}_2(0) = \mathsf{seed} :$$

the origin is not seed material of label 1 — it lies in the recorded bounded complementary chamber of S_1, the open unit disk — while for label 2 it lies in the seed itself. See Figure 7.

By monotonicity of Fill, the witness persists under *every* same-label enlargement $S_\ell \subset S'_\ell$: the enlarged completions satisfy $K_1 \subset K'_1$ and $K_2 \subset K'_2$, so $0 \in K'_1 \cap K'_2$ continues to witness a merger. The persistence clause of Theorem 9.3 is therefore sharp: the defect cannot be repaired by growing the seeds. What can remove it is only a revision outside the fixed assignment. For example, if the frozen packet input is revised to include the connecting radius $[\frac{1}{4}, 1]$ and all three pieces are assigned one label, the resulting seed is connected and its single completion is the closed unit disk, so there is no inter-packet merger. This changes both the input inventory and its typed assignment; it is not a repair by same-label enlargement of the two original packets. Within the fixed assignment, the failure certificate is final.

This is exactly why clause (3) of Definition 8.1 requires disjointness of the *completed* packets, not of the raw seeds. Raw-seed disjointness is not stable under fullification: an audit performed at seed level would certify the present system and hand Theorem 8.4 a family that is not pairwise disjoint, invalidating the duality argument of Lemma 8.3 at its first step. Auditing the completions, by contrast, costs nothing extra — Lemma 9.1 makes the completions canonical — and it converts every failure into a persistent witness (25).

9.6. **The cellwise certificate.** The audit clauses of Theorem 9.3 still require computing each completion and comparing every pair of completions across the localization. The final result of this section decomposes even this: it confines each packet to its own closed Jordan cell, inside which the completion is trapped automatically and the port verification becomes local to the cell. Disjointness of the completions then follows from disjointness of the cells, which is checked on N disks rather than on $\binom{N}{2}$ pairs of filled hulls.

Proposition 9.5 (Cellwise packet-and-port certificate; manuscript Proposition 10.15)**.** *Let $N \geq s \geq 1$, let U be a Jordan domain, and let $D_1, \dots, D_N \Subset U$ be pairwise disjoint closed Jordan disks. Index the atom packets first. Suppose every packet has a nonempty compact connected raw seed $S_\ell \Subset D_\ell$, the seeds cover the complete frozen inventory, $\mathcal{B}_i \subset S_i$ for $1 \leq i \leq s$, and every original atom A_i has an unprotected point $x_i \in A_i \setminus P_i$ and a Jordan arc $\gamma_i \colon [0,1] \to D_i$ satisfying*

$$\gamma_i(0) = x_i, \qquad \gamma_i\big((0,1)\big) \subset D_i \setminus S_i, \qquad \gamma_i(1) \in \partial D_i \tag{28}$$

(condition (9.11a) of [12]*). Then $K_\ell = \mathrm{Fill}(S_\ell) \Subset D_\ell$, the completed packets are pairwise disjoint, every $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$, and the packet system has simultaneous ports and one common free component; hence it supplies a protected ribbon-routing certificate and its realized carrier.*

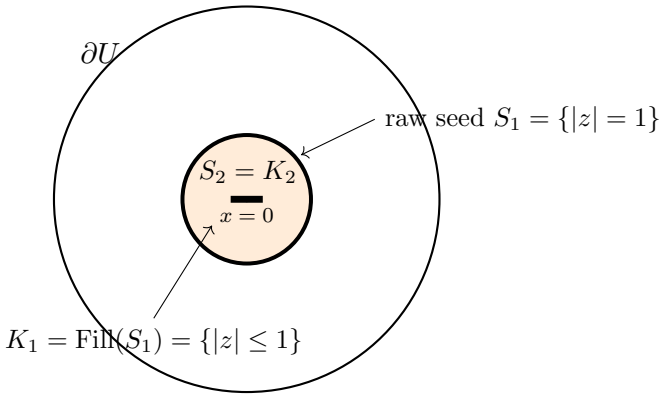


FIGURE 7. The post-fullification merger of Example 9.4 in the localization $U = \{|z| < 3\}$. The raw seeds — the circle S_1 and the segment S_2 — are disjoint, so every seed-level disjointness audit passes. But the completion $K_1 = \mathrm{Fill}(S_1)$ is the closed unit disk (shaded), which contains $S_2 = K_2$: the completed packets merge. The witness point $x = 0$ carries the record $(1, 2, 0, \mathrm{ch}_1(0), \mathrm{ch}_2(0))$ with $\mathrm{ch}_1(0)$ the bounded complementary chamber $\{|z| < 1\}$ of S_1 and $\mathrm{ch}_2(0) = \mathsf{seed}$. By monotonicity of Fill the merger persists under every same-label enlargement of the seeds; only a different typed assignment removes it.

Proof. Step 1: the completion is trapped in its cell. Fix ℓ. By the Schoenflies theorem, $S^2 \setminus D_\ell$ is an open Jordan disk containing ∞, so its planar part $\mathbb{C} \setminus D_\ell$ — an open disk with one point removed — is connected, and it is unbounded. Since $S_\ell \Subset D_\ell$ means $S_\ell \subset \operatorname{int} D_\ell$, the exterior $\mathbb{C} \setminus D_\ell$ is a connected unbounded subset of $\mathbb{C} \setminus S_\ell$ and therefore lies in the unbounded component $\Omega_\infty(S_\ell)$. Consequently every bounded component B of $\mathbb{C} \setminus S_\ell$ is disjoint from $\mathbb{C} \setminus D_\ell$ — distinct components of $\mathbb{C} \setminus S_\ell$ are disjoint — so $B \subset D_\ell$, and $\mathrm{Fill}(S_\ell) \subset D_\ell$. The containment is compact: a point $x \in \partial D_\ell$ does not lie in $S_\ell \subset \operatorname{int} D_\ell$, and every neighborhood of x meets $\mathbb{C} \setminus D_\ell \subset \Omega_\infty(S_\ell)$; since the component of $\mathbb{C} \setminus S_\ell$ containing x is open and contains a small ball around x, that component meets $\Omega_\infty(S_\ell)$ and hence equals it, so $x \in \Omega_\infty(S_\ell)$ and $x \notin \mathrm{Fill}(S_\ell)$. Thus $K_\ell = \mathrm{Fill}(S_\ell) \subset \operatorname{int} D_\ell$, and being compact, $K_\ell \Subset D_\ell$. The cells are pairwise disjoint, so the completions are pairwise disjoint, and $D_\ell \Subset U$ gives $K_\ell \Subset U$.

Step 2: the arc condition certifies exposure. Fix i. The point $\gamma_i(1)$ lies on the Jordan curve ∂D_i, which bounds the closed Jordan disk $S^2 \setminus \operatorname{int} D_i$ containing ∞ in its interior. Straightening this disk by the Schoenflies theorem [13] and taking the preimage of a radius, we obtain a Jordan arc β from $\gamma_i(1)$ to ∞ meeting D_i only at $\gamma_i(1)$. Consider the concatenated set

$$\Gamma \;:=\; \gamma_i\big((0,1]\big) \,\cup\, \big(\beta \setminus \{\infty\}\big),$$

a connected unbounded subset of $\mathbb{C}$. It is disjoint from S_i: the piece $\gamma_i((0,1))$ lies in $D_i \setminus S_i$ by (28); the point $\gamma_i(1)$ lies on ∂D_i, which misses $S_i \subset \operatorname{int} D_i$; and $\beta \setminus \{\gamma_i(1), \infty\}$ lies in $\mathbb{C} \setminus D_i \subset \mathbb{C} \setminus S_i$. A connected unbounded subset of $\mathbb{C} \setminus S_i$ lies in the unbounded component $\Omega_\infty(S_i)$, so in particular

$$\gamma_i\big((0,1]\big) \;\subset\; \Omega_\infty(S_i) \;=\; \mathbb{C} \setminus \operatorname{Fill}(S_i),$$

that is, $\gamma_i((0,1])$ misses K_i. Since $x_i \in A_i \subset S_i \subset K_i$, this gives

$$\gamma_i \cap K_i \;=\; \{x_i\}.$$

The arc γ_i maps into $D_i \subset U$, and $x_i \in A_i \setminus P_i$ by hypothesis, so γ_i is precisely an arc of the form read by (26): every $x_i \in \operatorname{Exp}(A_i, K_i; P_i)$.

Step 3: assembly. The three audit clauses of Theorem 9.3 now hold: clause (1), because the seeds are nonempty, compact, connected, and cover the complete frozen inventory by hypothesis; clause (2), by Step 1; and clause (3), by Step 2. Lemma 9.2 turns the individual arcs into pairwise disjoint simultaneous port arcs with positive-parameter parts in the common complement of the completions, and Theorem 9.3 certifies that the geometric augmented-packet clauses of Definition 8.1 hold for this assignment. Theorem 8.4 then determines one common free component of Ω_{free} with simultaneous singleton ports to every original atom, hence a protected ribbon-routing certificate and its realized carrier. □

This proposition is sufficient only, and we close the section by marking the boundary of what it proves, in the discipline of Section 10. The cell-wise certificate solves the planar verification problem once the cells D_ℓ, the complete seeds S_ℓ, and the unprotected arcs γ_i are supplied. The actual virtual theorem contemplated by the source manuscript must still *construct* those cells, complete seeds, and unprotected arcs, and must transport them through the final Wave horizon; that construction is posed in [12] only as an open problem, it is not addressed by anything in this paper, and no clause of Theorem 9.3 or Proposition 9.5 is promoted here to a statement about virtual carriers, the Wave horizon, or Mandelbrot local connectivity.

10. Scope, interfaces, and next gates

10.1. **What the finite theory closes.** For a completely specified finite inventory, the paper closes the planar routing problem at three levels.

(1) A protected ribbon-routing certificate is realized by a full carrier, with the declared prime-end chains, distinctness, and cyclic order preserved (Theorem 5.4).
(2) Existence of such a certificate is equivalent to one component of the frozen free domain having simultaneous singleton ports (Theorem 7.3).
(3) Theorem 9.3 gives each fixed packet assignment a finite audit with three monotone-persistent failure modes. Theorem 8.4 and Proposition 9.5 provide positive sufficient tests.

The attachment ledger additionally decides when replacing the certificate tree by a more general continuum connector preserves fullness (Theorem 6.1 and Corollary 6.3).

These conclusions are invariant under orientation-preserving conformal changes of the Jordan localization. They require no local connectivity of the atoms, no boundary parametrization, no presumed ray landing, and no metric width estimate.

10.2. **What remains outside the theorem.** The finite theory does not supply the data it audits. In particular it does not construct:

(1) a forward-coherent protected inventory at all depths;
(2) a virtual carrier or a Wave horizon;
(3) an analytic correspondence, covering action, or immersion action on such a horizon;
(4) a uniform incidence bound, marked-width estimate, or compression mechanism; or
(5) a proof of local connectivity for any new Mandelbrot parameter.

The direct-decoration rooted-end compactification and its finite carrier application belong to the companion paper [16]; importing that application here would duplicate its principal results without closing any of the infinite gates above.

10.3. **Three next research gates.** The results isolate three concrete directions in which a future argument can advance.

The inverse-envelope gate. Theorem 6.6 shows that decreasing full connector envelopes with connected traces pass safely to the limit. A dynamical construction of such envelopes, compatible with pullback and with one frozen family of relative collar tails, would bypass the two Hausdorff defects of Warning 6.7. What is missing is the construction, not another compactness extraction.

The quantitative routing gate. The exact free-component criterion is qualitative. It neither bounds the number of sides of the ribbon tree nor supplies a modulus, width, or incidence estimate. Any application requiring analytic compression must augment a topological certificate by a quantitative routing theorem inside the same frozen component. The separation ledger identifies the invariant that such a theorem must preserve, but does not provide the estimate.

The coherent-marking gate. Prime-end survival here is finite-family survival. Passing to infinitely many accesses requires a coherent choice of null chains and relative collar tails whose truncations remain compatible across levels. Pairwise distinctness at every finite stage is insufficient by itself: impressions may overlap in the atoms, and collar widths may collapse. A successful coherence theorem must therefore be formulated in the complement, in terms of relative tails and selected sides, rather than as pointwise separation of impressions.

These gates are intentionally stated as proof obligations, not conjectural consequences of the present paper. The value of the finite theory is that a future construction can now be tested against exact planar conditions, and a failure produces a persistent witness rather than an ambiguous instruction to "choose a better hull."

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Appendix A. Provenance and statement crosswalk

The certificate, criterion, and packet results originated in the source research manuscript [12]. This draft corrects the slit statement, uses null crosscut chains and relative collar tails throughout, repairs the single-atom construction, and adds the separation ledger, the sharp Hausdorff-limit example, and nested-limit stability. No statement labeled open or conditional in the source has been promoted.

Present result	Source	Status in this paper
Theorem 3.1	Theorem 6.12	extracted
Proposition 3.2	Proposition 6.15	extracted
Definition 5.1	Definition 10.1	relative-tail and incidence conventions made explicit
Lemma 5.2	Lemma 10.2	two proofs supplied
Lemma 5.3	Lemma 10.3	relative-tail formulation
Theorem 5.4	Theorem 10.4	proof synchronized with corrected conventions
Proposition 5.6	Proposition 10.5	extracted
Theorem 6.1	Draft II	classical duality/Mayer–Vietoris consequence, packaged for carriers
Corollary 6.3	Draft II	classical zero-defect consequence, packaged for carriers
Proposition 6.5	Draft II	Hausdorff-limit separation example proved in this draft
Theorem 6.6	Draft II	decreasing-envelope theorem proved in this draft
Lemma 7.1	Lemma 10.7	repaired; see Remark 7.2
Theorem 7.3	Theorem 10.8	proof rebuilt around the repaired slit lemma; $s = 1$ included
Definition 8.1	Definition 10.9	compact atom–tail blocks and named compact obstacle pieces
Lemma 8.3	Lemma 10.10	compacta duality made explicit
Theorem 8.4	Theorem 10.11	extracted
Lemma 9.1	Lemma 10.12	extracted
Lemma 9.2	Lemma 10.13	extracted
Theorem 9.3	Theorem 10.14	extracted
Proposition 9.5	Proposition 10.15	extracted

Examples 7.4 and 9.4, together with Section 4, are expository additions. The direct-decoration application formerly included in Draft I has been removed because its strengthened form is the subject of the companion paper [16].

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